Fully-Dualizable and Invertible $\mathcal{E}_n$-Algebras

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January 14, 2026

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1 Introduction

One of the central insights in the modern study of topological quantum field theory (TQFT) is that the structure of a field theory is governed by higher-categorical dualizability conditions. This idea originates in the pioneering work of Baez-Dolan [BD95], who formulated the Cobordism Hypothesis. They propose that an n-dimensional fully extended TQFT should be the data of: a symmetric monoïdal (∞, m)-category $\mathcal{A}$, and a "fully-n-dualizable" object $a \in \mathcal{A}$. The initial such TQFT is then conjectured to be given by the (∞, n)-category of framed bordisms. Typically, the symmetric monoïdal (∞, m)-categories $\mathcal{A}$ of interest arise from studying algebraic structures and the Cobordism Hypothesis can be seen as a duality between algebra and manifold topology.

Lurie observed in [Lur09, Sec. 4.1] that the collection of $\mathcal{E}_n$-algebras, in some fixed symmetric monoïdal $(\infty, 1)$-category $\mathcal{V}$ with compatible colimits, can be arranged into such a symmetric monoïdal $(\infty, n+1)$-category $\mathbf{Mor}_n(\mathcal{V})$ called the higher Morita category. This generalises the well-known classical Morita $(2, 2)$-category whose objects are non-commutive rings and whose morphisms are bimodules. The idea of this category was first sketched by Lurie in a [Lur09, Def. 4.1.11] and later formalized by Scheimbauer and Haugseng in [Sch14, Hau17a, Hau23]. It's clear from the construction that all k-morphisms in $\mathbf{Mor}_n(\mathcal{V})$, for $1 \leq k < n$, have both left and right adjoints, implying that any $\mathcal{E}_n$-algebras A automatically gives rise to a n-dimensional fully-extended TQFT. The computations of the partition function of this TQFT is exactly given by factorization homology with coefficients in A as predicted by Lurie [Lur09, Thm. 4.1.24] and formalized in [Sch14]. The n-morphisms of $\mathbf{Mor}_n(\mathcal{V})$ do not however necessarily have adjoints, which invites the question of characterizing those $\mathcal{E}_n$-algebras A for which the associated TQFT extends to $(n+1)$-dimensions, i.e., the fully-dualizable $\mathcal{E}_n$-algebras. The purpose of this paper is to answer this question, and further classify which $\mathcal{E}_n$-algebras form an invertible TQFT.

Classically, the fully-dualizable objects of the classical Morita $(2, 2)$-category are known as smooth and proper algebras, and the invertible objects are known as Azumaya algebras. These notions were generlized to the dg-setting by Toën [Toë12]. More explicitly, an algebra A in ${}_k\mathbf{Mod}$ is smooth if A is perfect as an $A^{op} \otimes_k A$-module, and proper if A is perfect as a k-module. A smooth and proper algebra A is further Azumaya if the canonical

$$k \xrightarrow{\sim} \mathbf{HC}_{\mathcal{E}^1}(A)$$
$$A^{\mathrm{op}} \otimes_k A \xrightarrow{\sim} \mathbf{End}_k(A)$$

are equivalences.

Lurie claimed in [Lur09, Rem. 4.1.27] that unwinding the proof of the Cobordism Hypothesis should give an explicit generalization of the above conditions for full-dualizability of $\mathcal{E}_n$-algebras, involving factorization homology. This criterion was later stated as a conjecture by Brochier, Jordan, Safronov, Snyder [BJSS21, Conj. 1.8], which we prove in more generality in Theorem A stated below.

Theorem 1.1 (Thm. A)

Let $0 \leq k < n$ and $\mathcal{V} \in \mathbf{Alg}_{\mathcal{E}^\infty}(\mathbf{Cat}_{(\infty,1)})$ whose tensor product commutes with sifted colimits in both variables. Let M be a k-morphism in $\mathbf{Mor}_n(\mathcal{V})$. Then M is fully-dualizable if and only if, for all $0 \leq i \leq n-k$, the canonical $\mathcal{E}_1$-actions

$$\int_{(D^i, S^{i-1})} (A, M) \curvearrowright M \curvearrowleft \int_{(D^i, S^{i-1})} (B, M)$$

both exhibit M as a dualizable module, where A (resp. B) are either the (co)domain of M if $1 \leq k$, or the identity $\mathbb{1}_{\mathcal{V}}$ if $k = 0$.

Brochier *et al.* also conjectured a generalization of the characterization of Azumaya algebras [BJSS21, Conj. 1.9], which we prove in in Theorem B stated below.

Theorem 1.2 (Thm. B)

Let $0 \leq k < n$ and $\mathcal{V} \in \mathbf{Alg}_{\mathcal{E}^\infty}(\mathbf{Cat}_{(\infty,1)}))$. Let M be a k-morphism in $\mathbf{Mor}_n(\mathcal{V})$. Then M is invertible if and only if it is fully-dualizable and, for all $1 \leq i \leq n-k$, the universal maps

$$\int_{(D^i, S^{i-1})} (A, M) \xrightarrow{\sim} \mathbf{HC}^B_{\mathcal{E}^{i-1}}(M)$$
$$\int_{(D^i, S^{i-1})} (B, M) \xrightarrow{\sim} \mathbf{HC}^A_{\mathcal{E}^{i-1}}(M)$$

are invertible, where A (resp. B) are either the (co)domain of M if $1 \leq k$, or the identity $\mathbb{1}_{\mathcal{V}}$ if $k = 0$.

1.1 Overview

Section 2 In order to establish these results, we first need to prove a few preliminary results about (∞, n)-categories with adjoints in section 2. Most of these results are well-known to experts, but lacking of any rigorous proof in the literature, so we include them here for completeness. All of the work of this section heavily relies on the seminal result of Riehl and Verity [RV16, Thm. 4.4.11] bringing adjunctions in $(\infty, 2)$-categories into a powerful homotopy coherent framework. One of the most important of these folklore theorem is that the inclusion $\mathbf{Cat}^{\mathrm{adj}}_{(\infty,n)} \hookrightarrow \mathbf{Cat}_{(\infty,n)}$ has left and right adjoints (see Lemma 2.8). The existence of the left adjoint is trivial from the definition, but the existence of the right adjoint is more subtle. One can construct it explicitly by hand, realizing the idea that it ought to be given by the full sub-(∞, n)-category on these morphisms which are fully-adjointable. We choose here to construct this functor more formally, using the Adjoint Functor Theorem [Lur18, Thm. 06Q5] by proving that $\mathbf{Cat}^{\mathrm{adj}}_{(\infty,n)} \hookrightarrow \mathbf{Cat}_{(\infty,n)}$ preserves colimits:

Proposition 1.3 (Lemma 2.6)

Let $F\colon I \to \mathbf{Cat}_{(\infty,n)}$ be a diagram of (∞, n)-categories which all admit right (resp. left) adjoints for all k-morphisms. Then so does $\varinjlim F$.

The argument relies on the explicit way in which colimits of (∞, n)-categories are constructed. More explicitly, a k-morphism in some $\varinjlim_{i\in\mathcal{I}} \mathcal{C}_i$ is always given by some intricate pasting of k-morphisms in the $\mathcal{C}_i$'s. This is an intrinsic consequence of the "Segalification" construction for (∞, n)-categories described in [BS24]. It's then well-known that composition of adjoints should have adjoints, which will in turn imply that $\mathbf{Cat}^{\mathrm{adj}}_{(\infty,n)}$ is stable under colimits. The main advantage of this approach is that it gives a general method for proving that some category, defined as a colimit, has adjoints. The author sees this as a powerful tool to show results of the type: "to construct a fully-adjoint/dualizable thingy, one only needs to exhibit this and that (co)unit". A first statement of the type is given by the result bellow:

Proposition 1.4 (Lemma 2.10)

We have a colimit diagram

$$\begin{tikzcd}
 & & & \mathsf{Free}^{\{k\}}(c_k) & & & \\
\cdots \arrow[rrr, dashed, two heads] & & & & & & \cdots \arrow[lll, dashed, two heads] \\
 & \mathbf{Adj}_k \arrow[uul, dashed, two heads] & & \mathbf{Adj}_k \arrow[uu, dashed, two heads] & & \mathbf{Adj}_k \arrow[uur, dashed, two heads] & \\
 & c_k \arrow[ul, two heads, "R"] \arrow[ur, two heads, "L"'] & & c_k \arrow[ul, two heads, "R"] \arrow[ur, two heads, "L"'] & & c_k \arrow[ul, two heads, "R"] \arrow[ur, two heads, "L"'] & & c_k \arrow[ul, two heads, "R"] \arrow[ur, two heads, "L"'] &
\end{tikzcd}$$

where all projections are epimorphisms.

In essence, this can be seen as formalizing the idea: "to give a fully-adjoint 1-morphism is equivalent to providing a sequence of candidates for the iterated adjoints and for the (co)units". The proof of the main Theorem relies on a similar guiding principle: "to show f is fully-adjoint, it suffices to find adjoints for the iterated (co)units of the form $\eta(\epsilon(\epsilon(\ldots\eta(f)\ldots)))$" (where we range over all words in ϵ and η of some fixed length).

Section 3 In the following section, we then focus on the geometric and algebraic tools required for our proofs. On the geometric side, the main ingredient here is the powerful theory of factorization homology of conically smooth stratified spaces developed by Ayala, Francis, Rozenblyum and Tanaka in [AFT17b, AFR18b, AFT17a]. For us, factorization homology is the natural tool to manipulate these $\mathcal{E}_k$-algebras in iterated bimodules M. Following the guiding principle above, factorization homology will allow us to explicitly describe the algeraic structure corresponding to each of these iterated (co)units $\epsilon(\eta(\epsilon(\ldots\eta(M)\ldots)))$. We define various stratified spaces which will "carry" this dualizability data, and various stratified map which induces functors sending "M" to these "$\eta(\eta(\epsilon(\ldots\epsilon(M)\ldots)))$". These spaces can be explicitly visualized as stratifications of the usual handlebodies. Their algebraic structure can be visualized as iteratively "pinching" these handlebodies, through constructible bundle maps.

Finally, we prove a couple technical algebraic results. One of the main ingredient in our proof of Theorem A is a criterion (Lemma 3.6) for when n-morphisms in $\mathbf{Mor}_n(\mathcal{V})$ have adjoints, essentially generalizing [Lur09, Prop. 4.6.2.13]. Finally, to complete our proof of Theorem B, we also need to understand how Hoschild cohomology appears as duality data in $\mathbf{Mor}_n(\mathcal{V})$, which is done in Lemma 3.9.

Section 4 This short section is the compiled proofs of both main Theorems.

1.2 Acknowledgments

The author would like to thank his PhD supervisor John Francis for many helpful discussions.

1.3 Notations

- For $n \geq 0$, we denote by $\mathbf{Cat}_{(\infty,n)}$ the presentable $(\infty,1)$-category of (∞,n)-categories. We have a sequence of reflections and coreflections:

$$\begin{tikzcd}
\mathcal{S} \arrow[r, hook] & \mathbf{Cat}_{(\infty,1)} \arrow[l, bend right, "|-|_{>0}"'] \arrow[l, bend left, "-_{\leq 0}"] \arrow[r, hook] & \mathbf{Cat}_{(\infty,2)} \arrow[l, bend right, "|-|_{>1}"'] \arrow[l, bend left, "-_{\leq 1}"] \arrow[r, hook] & \cdots \arrow[l, bend right, "|-|_{>2}"'] \arrow[l, bend left, "-_{\leq 2}"] \arrow[r, hook] & \mathbf{Cat}_{(\infty,n)} \arrow[l, bend right, "|-|_{>n-1}"'] \arrow[l, bend left, "-_{\leq n-1}"] \arrow[r, hook] & \cdots \arrow[l, bend right, "|-|_{>n}"'] \arrow[l, bend left, "-_{\leq 0}"]
\end{tikzcd}$$

- For $n \geq 0$, we have an adjunction

$$\begin{tikzcd}
\mathbf{Cat}_{(n,n)} \arrow[r, hook] & \mathbf{Cat}_{(\infty,n)} \arrow[l, bend right, "\mathcal{H}o_n"']
\end{tikzcd}$$

- We denote by c_k the walking k-morphism, which we can see as an (∞,n)-category for all $n \geq k$. We denote by ∂c_k the boundary of the walking k-morphism, i.e., the walking pair of parallel $(k-1)$-morphisms, which can also be seen as an (∞,n)-category for all $n \geq k-1$. It's the colimit of the diagram

$$\begin{tikzcd}
c_0 \arrow[r] \arrow[dr] \arrow[drr, dashed] \arrow[drrr, dashed] & c_1 \arrow[r] \arrow[dr] \arrow[drr, dashed] & \cdots \arrow[r] \arrow[dr] & c_{k-1} \arrow[dr, dashed] & \\
& & & & \partial c_k \\
c_0 \arrow[r] \arrow[ur] \arrow[urrr, dashed] & c_1 \arrow[r] \arrow[ur] \arrow[urr, dashed] & \cdots \arrow[r] \arrow[ur] & c_{k-1} \arrow[ur, dashed] &
\end{tikzcd}$$

  A k-morphism $f\colon c_k \to \mathcal{C}$ is also denoted $f\colon a_{k-1} \to b_{k-1}\colon \ldots \colon a_0 \to b_0\colon \mathcal{C}$ where $a_i, b_i\colon c_i \to \mathcal{C}$ are the induced projections above.

- A refinement of stratified space will always be denoted by a dashed arrow $X \dashrightarrow Y$. A (weak) constructible bundle will always be denoted by a double-headed arrow $X \twoheadrightarrow Y$. An open embedding will always be denoted by a hooked arrow $X \hookrightarrow Y$. A closed inclusion will always be denoted by a hooked arrow $X \overset{\text{cl}}{\hookrightarrow} Y$ with the label "cl".

2 Adjoints in (∞, n)-categories

Definition 2.1 (Adjoint)**.** Let $n > k \geq 1$ and $\mathcal{C}\colon \mathbf{Cat}_{(\infty,n)}$. We say that $l\colon a_{k-1} \to b_{k-1}\colon \ldots : a_0 \to b_0 \colon \mathcal{C}$ is a left adjoint if its equivalence class in $\mathcal{H}o_2(\mathcal{C}(a_0, b_0)(a_1, b_1)\ldots(a_{k-2}, b_{k-2}))$ is a left adjoint.

Construction 2.1 (Free/Walking Adjunction)*.* Recall that Schanuel and Street [SS86] defined combinatorially the free $(2, 2)$-category **Adj** containing an adjunction $L\colon\ + \underset{\longleftarrow}{\overset{\longrightarrow}{\perp}} - \ : R$. This category was later studied in depth by Riehl and Verity [RV16], in a higher categorical context.

The results of this paper rely greatly on the celebrated result from [RV16], recalled bellow.

Theorem 2.1 (Homotopy Uniqueness of Adjunctions [RV16, Thm. 4.4.11])

Let $\mathcal{C}\colon \mathbf{Cat}_{(\infty,2)}$. Precomposition by $L, R\colon c_k \to \mathbf{Adj}$ gives monomorphisms

$$[\mathbf{Adj}, \mathcal{C}] \underset{R^*}{\overset{L^*}{\rightrightarrows}} [c_1, \mathcal{C}]$$

selecting the connected components of left (resp. right) adjoint morphisms.

2.1 Colimits of (∞, n)-categories with adjoints have adjoints

In this section, we prove that a colimit of a diagram of (∞, n)-categories are given as compositions of k-cells in the categories in the diagram. For this purpose, we use the following equivalent definitions of of (∞, n)-categories as:

- Rezk's complete Segal Θ_n-spaces [Rez10], inspired by Joyal's unpublished ideas [Joy97],
- Barwick's n-fold complete Segal spaces [Bar05], inspired by Rezk's complete segal spaces (for $n = 1$) [Rez01],

In [BSP20], Barwick and Schommer-Pries proved that these two definitions are equivalent. In [Hau17b], Haugseng further proved that the two theories coincide priori to univalent completion. Combining both results, we get the diagram where left (resp. right) adjoints commute with the vertical equivalences, and where the last column is equivalent to $\mathbf{Cat}_{(\infty,n)}$.

$$\begin{tikzcd}
\mathcal{P}\mathrm{sh}(\Delta^n) \arrow[r, bend left, "\mathbb{L}_S"] \arrow[r, phantom, "\perp"] & \mathbf{Seg}(\Delta^n) \arrow[l, bend left] \arrow[d, "\wr"] \arrow[r, bend left, "\mathbb{L}_C^{\Delta}"] \arrow[r, phantom, "\perp"] & \mathbf{Seg}^{\mathrm{univ}}(\Delta^n) \arrow[l, bend left] \arrow[d, "\wr"] \\
& \mathbf{Seg}(\Theta^n) \arrow[r, bend left, "\mathbb{L}_C^{\Theta}"] \arrow[r, phantom, "\perp"] & \mathbf{Seg}^{\mathrm{univ}}(\Theta^n) \arrow[l, bend left]
\end{tikzcd} \tag{2.1}$$

Note that an explicit formula for the Δ^n-Segalification $\mathbb{L}_S^{\Delta}$ was established in [BS24, Thm. D] by Barkan and Steinebrunner following the theory of necklaces developped by Dugger and Spivak [DS11]. Another formula was established independently for $n = 1$ by Ayala and Francis [AF24, Sec. 3]. An explicit formula for Θ^n-univalent completion $\mathbb{L}_C^{\Theta}$ was established in [AF18, Prop. 2.4], generalizing the Rezk's original formula [Rez01]. Using these formulas and using that left adjoints preserves colimits, we can understand explicitly how the k-cells of a colimit of categories are built from the k-cells of the categories in the diagram.

Construction 2.2 (Suspension)*.* Recall that we have a functor $\sigma\colon \mathcal{P}\mathrm{sh}(\Delta^n) \to \mathcal{P}\mathrm{sh}(\Delta^{n+1})$ defined as follows:

$$\sigma X_{a_1,\ldots,a_{n+1}} = \{\bullet_-\} \sqcup \coprod_{i=1}^{a_1} X_{a_2,\ldots,a_{n+1}} \sqcup \{\bullet_+\}$$

where the maps are defined in the obvious way. Note that if X is reduced/Segal (in the ith direction)/univalent, then σX is reduced/Segal (in the $(i+1)^{\text{th}}$ direction)/univalent. We can iterate this construction and get

$$\sigma^k X_{a_1,\ldots,a_{n+k}} = (\partial c_k)_{a_1,\ldots,a_{k-1},0,\ldots,0} \coprod \left(X_{a_{k+1},\ldots,a_{n+k}} \times \prod_{i=1}^{k} [a_i] \right)$$

Note that $\sigma^k c_l = c_{l+k}$. We also clearly have an adjunction

$$\begin{tikzcd}
\mathbf{Cat}_{(\infty,n)} \arrow[r, bend right, "\sigma^k"'] \arrow[r, phantom, "\top"] & \partial c_k \backslash \mathbf{Cat}_{(\infty,n+k)} \arrow[l, bend right, "\hom"']
\end{tikzcd}$$

where for $a_{k-1}, b_{k-1}\colon \ldots : a_0 \to b_0 \colon \mathcal{C}$, we denote the corresponding $\hom(\partial c_k \to \mathcal{C}) = \mathcal{C}(a_0, b_0)(a_1, b_1)\ldots(a_{k-2}, b_{k-2})$.

We denote by $\mathbf{Adj}_k = \sigma^{k-1}\mathbf{Adj}$ the free $(\infty, k+1)$-category containing an adjoint k-morphism. We have an immediate corollary of Lemma 2.1.

Corollary 2.1.1 (Homotopy Uniqueness of (higher) Adjunctions [RV16, Thm. 4.4.11])

Let $1 \leq k < n$ and $\mathcal{C}\colon \mathbf{Cat}_{(\infty,n)}$. Precomposition by $L, R\colon c_k \to \mathbf{Adj}_k$ gives monomorphisms

$$\begin{tikzcd} {[\mathbf{Adj}_k, \mathcal{C}]} \arrow[r, hook, bend left, "L^*"] \arrow[r, hook, bend right, "R^*"'] & {[c_k, \mathcal{C}]} \end{tikzcd}$$

selecting the connected components of left (resp. right) adjoint k-morphisms.

Proof: σ^{k-1} is a left adjoint composed with a forgetful from a slice, so it preserves contractible colimits and thus epimorphisms. Also, the notion of adjoint k-morphism is clearly a suspension of that of adjoint 1-morphisms. ■

To formulate the next few lemmas, we now extend the notion of "having adjoints" to n-fold simplicial spaces. Note that the homotopy uniqueness results above do not apply to general n-fold simplicial spaces, only to Segal and univalent/complete ones. In other words, $c_k \to \mathbf{Adj}_k$ is not an epimorphism in $\mathcal{P}\mathrm{sh}(\Delta^n)$ or $\mathbf{Seg}(\Delta^n)$ for $k < n$.

Definition 2.2 (n-Fold Simplicial Space with Adjoints)**.** Let $1 \leq k < n$. Let $X \in \mathcal{P}\mathrm{sh}(\Delta^n)$ be a reduced n-fold simplicial space. We say that X has right (resp. left) adjoints for all k-cells if for all $c_k \to X$ for which there exists a (not necessarily unique) lift

$$\begin{tikzcd} c_k \arrow[r] \arrow[d] & X \\ \mathbf{Adj}_k \arrow[ur, dashed] & \end{tikzcd}$$

where the downward functor is L (resp. R).

Recall from [BS24, Thm. D] that, for X reduced, the Segalification functor can be computed as $\mathbb{L}^{\Delta}_{S} = \mathbb{L}^{\Delta}_{S,n} \circ \ldots \circ \mathbb{L}^{\Delta}_{S,1}$.

Lemma 2.2

Let $1 \leq i$ and X be a n-fold Segal space. Then $\mathbb{L}^{\Delta}_{S,i}\left(\sigma^i X \coprod_{c_{i-1}} \sigma^i X\right)$ is an $(n+i+1)$-fold Segal space.

Proof: It suffices to prove the result for $i = 1$ as Segalification clearly commutes with suspension. By Remark 2.2, we have

$$\left(\sigma X \coprod_{c_0} \sigma X\right)_{a_1,\ldots,a_{n+1}} = \{\bullet_0\} \sqcup \coprod_{i=1}^{a_1} X_{a_2,\ldots,a_{n+1}} \sqcup \{\bullet_1\} \sqcup \coprod_{i=1}^{a_1} X_{a_2,\ldots,a_{n+1}} \sqcup \{\bullet_2\}$$

By inspection, using the Necklace formula [BS24, Thm. A], we get

$$\left(\mathbb{L}^{\Delta}_{S,i}\left(\sigma X \coprod_{c_0} \sigma X\right)\right)_{a_1,\ldots,a_n} = \{\bullet_0\} \sqcup \coprod_{i=1}^{a_1} X_{a_2,\ldots,a_{n+1}} \sqcup \{\bullet_1\} \sqcup \coprod_{i=1}^{a_1} X_{a_2,\ldots,a_{n+1}} \sqcup \{\bullet_2\} \coprod_{1 \leq i < j \leq a_1} X^2_{a_2,\ldots,a_{n+1}}$$

This is easily seen to be Segal. ■

Corollary 2.2.1 (Composition of Adjoints)

Let $1 \leq i < k < n$. There are lifts

$$\begin{tikzcd} \mathbf{Adj}^k \arrow[d, dashed] & c_k \arrow[l, "L"'] \\ \mathbb{L}^{\Delta}_{S,i}\left(\mathbf{Adj}^k \coprod_{c_{i-1}} \mathbf{Adj}^k\right) & c_k \coprod_{c_{i-1}} c_k \arrow[l] \arrow[u] \end{tikzcd} \qquad \begin{tikzcd} \mathbf{Adj}^k \arrow[d, dashed] & c_k \arrow[l, "L"] \\ \mathbb{L}^{\Delta}_{S,k+1} \circ \mathbb{L}^{\Delta}_{S,k}\left(\mathbf{Adj}^k \coprod_{c_{k-1}} \mathbf{Adj}^k\right) & c_k \coprod_{c_{k-1}} c_k \arrow[l] \arrow[u] \end{tikzcd}$$

Proof: By Lemma 2.2 above, $\mathbb{L}^{\Delta}_{S,i}\left(\mathbf{Adj}^k \bigsqcup_{c_{i-1}} \mathbf{Adj}^k\right)$ is Segal. Note that $\mathbb{L}^{\Delta}_{S,k+1} \circ \mathbb{L}^{\Delta}_{S,k}\left(\mathbf{Adj}^k \bigsqcup_{c_{k-1}} \mathbf{Adj}^k\right)$ is also clearly Segal by Remark 2.2. The only equivalences in these n-fold simplicial sets are the identities, hence they are also complete. By Corollary 2.1.1, it then suffices to show in classical (strict) (n,n)-category theory that an ith-composition of k-adjoints is an adjoint. This is a standard result from ordinary category theory. ■

Lemma 2.3 (Segalification Preserves Adjoints)

Let X be a reduced n-fold simplicial space that has right (resp. left) adjoints for all k-cells. Then its Segalification $\mathbb{L}^{\Delta}_{S}(X)$ has right (resp. left) adjoints for all k-cells.

Proof: Using $\mathbb{L}^{\Delta}_{S} = \mathbb{L}^{\Delta}_{S,n} \circ \ldots \circ \mathbb{L}^{\Delta}_{S,1}$ for X reduced, and using [BS24, Lem. 2.15.a] which ensures X remains reduced through these steps, one can observe that it's sufficient to prove the result separately: for $\mathbb{L}^{\Delta}_{S,i}$ with $1 \leq i \leq k-1$, and for $\mathbb{L}^{\Delta}_{S,k+1} \circ \mathbb{L}^{\Delta}_{S,k}$.

- **Case $i < k$:** Fix $c_k = \Delta_{1,\ldots,1,0,\ldots,0} \to \mathbb{L}^{\Delta}_{S,i}(X)$. By the Necklace formula [BS24, Thm. A] and by reducedness, we can find $m \geq 1$ and a diagram

$$\begin{tikzcd}
c_k \arrow[r] & \Delta_{1,\ldots,1,m,1,\ldots,1,0,\ldots,0} \arrow[r] & \mathbb{L}^{\Delta}_{S,i}(X) \\
& c_k \bigsqcup_{c_{i-1}} \cdots \bigsqcup_{c_{i-1}} c_k \arrow[r] \arrow[u, "\simeq_i"'] & X \arrow[u]
\end{tikzcd}$$

where the arrows marked by $\simeq_i$ are Segal equivalences for the ith-coordinate Segal conditions. Without loss of generality, it suffices to consider the case $m = 2$. Using the universal property of $\mathbb{L}_{S,i}$ and using Corollary 2.2.1, we get a diagram

$$\begin{tikzcd}
c_k \arrow[rr, "L"] \arrow[d] & & \mathbf{Adj}_k \arrow[dl, "2.2.1"'] \arrow[dd, dashed] \\
\Delta_{1,\ldots,1,2,1,\ldots,1,0,\ldots,0} \arrow[r] \arrow[drr] & \mathbb{L}_{S,i}\left(\mathbf{Adj}_k \bigsqcup_{c_i} \mathbf{Adj}_k\right) \arrow[dr, dashed] & \\
c_k \bigsqcup_{c_i} c_k \arrow[u, "\simeq_i"] \arrow[r, "L \bigsqcup_{c_i} L"] \arrow[drr] & \mathbf{Adj}_k \bigsqcup_{c_i} \mathbf{Adj}_k \arrow[u, "\simeq_i"'] \arrow[dr, dashed] & \mathbb{L}^{\Delta}_{S,i}(X) \\
& & X \arrow[u, "\simeq_i"']
\end{tikzcd}$$

We want to show the top pentagone commutes. Because every other polytope commutes in the diagram, we have the desired lift.

- **Case $i = k$:** Fix $c_k = \Delta_{1,\ldots,1,0,\ldots,0} \to \mathbb{L}^{\Delta}_{S,k+1} \circ \mathbb{L}^{\Delta}_{S,k}(X)$. Because $\mathbb{L}^{\Delta}_{S,k+1}$ doesn't affect the space of k-morphisms, we can factor this map through $c_k = \Delta_{1,\ldots,1,0,\ldots,0} \to \mathbb{L}^{\Delta}_{S,k}(X)$. We then reproduce the same argument, keeping the first diagrm identical, and replacing $\mathbb{L}^{\Delta}_{S,i}$ by $\mathbb{L}^{\Delta}_{S,k+1} \circ \mathbb{L}^{\Delta}_{S,k}$ everywhere in the second diagram. ■

Lemma 2.4 (Partial Adjoints in Segalification)

Let X be a reduced n-fold simplicial space. Let $\mathcal{I} \subset \pi_0 X_{1,0,\ldots,0}$ be the minimal subset such that:

- *if $f\colon c_1 \to X$ has a right (resp. left) adjoint, as defined in Definition 2.2, f in $\pi_0 X_{1,0,\ldots,0}$ lies in $\mathcal{I}$,*
- *if $f\colon c_1 \to X$ can be written as a composition of morphisms in $\mathcal{I}$, then f lies in $\mathcal{I}$: in other words, if we have $\Delta_{r,0\ldots,0} \twoheadrightarrow c_1$ preserving end-points, and, if for all consecutive inclusions $c_1 \hookrightarrow \Delta_{r,0\ldots,0}$, the composotion lies in $\mathcal{I}$, then f lies in $\mathcal{I}$.*

$$\begin{tikzcd}
c_1 \arrow[d, "\forall"'] \arrow[dr, "\in\mathcal{I}"] & \\
\Delta_{r,0\ldots,0} \arrow[r] \arrow[d] & X \\
c_1 \arrow[ur, "f"'] &
\end{tikzcd}$$

Suppose $\mathcal{I} = \pi_0 X_{1,0,\ldots,0}$ is the whole set. Then $\mathbb{L}^{\Delta}_{S}(X)$ has right (resp. left) adjoints for all 1-cells.

Proof: The proof is almost identical to second part of the proof above. One can simply observe with the same diagram that any 1-cell in $\mathcal{I}$ has an adjoint in the Segalification. ■

Lemma 2.5 (Completion Preserves Adjoints)

Let X be an n-fold Segal space. Let $c_k \to \mathbb{L}_C X$ be a k-cell in the completion of X. There exists a k-cell of $c_k \to X$ factoring $c_k \to \mathbb{L}_C X$.

Proof: Note that by (2.1), one prove this result for Θ^n-spaces instead of n-fold Segal spaces. By [AF18, Prop. 2.4], we have an explicit formula for the univalent completion:

$$[c_k, \mathbb{L}_C X] = \varinjlim_{\Theta_n} [c_k \times E(-), X]$$

where $E\colon \Theta_n \to \mathcal{P}\mathrm{sh}(\Theta_n)$ sends the $T \in \Theta_n$ to the free strict n-groupoid generated by T (see [AF18, Def. 2.1] for more details). In particular, for any $c_k \to \mathbb{L}_C X$, we have a factorization

$$\begin{tikzcd}
 & c_k \arrow[r] & \mathbb{L}_C X \\
c_k \times E(\bullet) \arrow[ur, equal] \arrow[r, "\simeq_C"'] & c_k \times E(T) \arrow[u, "\simeq_C"'] \arrow[r] & X \arrow[u, "\simeq_C"']
\end{tikzcd}$$

■

We can now state the main result of this subsection:

Proposition 2.6 (Colimits have Adjoints)

Let $F\colon I \to \mathbf{Cat}_{(\infty,n)}$ be a diagram of (∞, n)-categories which all admit right (resp. left) adjoints for all k-morphisms. Then so does $\varinjlim F$.

Proof: By (2.1), one can compute the colimit as $\mathbb{L}_C \circ \mathbb{L}_S^{\Delta}\left(\varinjlim F\right)$ where the colimit is computed in $\mathcal{P}\mathrm{sh}(\Delta^n)$. Clearly, the collection of simplicial spaces with right (resp. left) adjoints for all k-morphisms is closed under colimits in $\mathcal{P}\mathrm{sh}(\Delta^n)$ by definition. By Lemma 2.3, $\mathbb{L}_S^{\Delta}\left(\varinjlim F\right)$ has right (resp. left) adjoints for all k-morphisms. Fix $c_k \to \mathbb{L}_C \circ \mathbb{L}_S^{\Delta}\left(\varinjlim F\right)$ a k-morphism. By Lemma 2.5, we have a factorization through $c_k \to \mathbb{L}_S^{\Delta}\left(\varinjlim F\right)$ which then has an adjoint. ■

2.2 Freely Adding Adjoints and Subcategories with Adjoints

Using the results of the previous subsection, we now establish some basic constructions of the (∞, n)-categories with adjoints. First, note that by Corollary 2.1.1, we find:

Corollary 2.6.1

Let $k \geq 0$. One can write $\mathbf{Adj}_k$ a finite colimit diagram of the c_i's.

Proof: Existence of L, R, η, ϵ and the commutativity of the triangle identities can easily be encoded by some finite diagram. The Theorem above shows that the canonical cocone on $\mathbf{Adj}_k$ that one gets is colimiting. ■

Definition 2.3 (Categories with Adjoints)**.** Let $S \subset \{1, \ldots, n\}$ and $\mathcal{A}\colon \mathbf{Cat}_{(\infty,n+1)}$. We say that $\mathcal{A}$ has S-adjoints if, for all $k \in S$ and all $c_k \to \mathcal{A}$, we have lifts

$$\begin{tikzcd}
\mathbf{Adj}_k \arrow[dr, dashed] & \\
c_k \arrow[u, "L"] \arrow[d, "R"'] \arrow[r] & \mathcal{A} \\
\mathbf{Adj}_k \arrow[ur, dashed] &
\end{tikzcd}$$

Denote $\mathbf{Cat}_{(\infty,n+1)}^{S\text{-adj}} \hookrightarrow \mathbf{Cat}_{(\infty,n+1)}$ the full subcategory of $(\infty, n+1)$-categories with S-adjoints.

Note that by the Theorem above, these lifts are unique if they exists.

Lemma 2.7 (Stability under Limits)

Let $S \subset K \subset \{1, \ldots, n\}$. The full subcategory $\mathbf{Cat}_{(\infty,n+1)}^{S\text{-adj}} \hookrightarrow \mathbf{Cat}_{(\infty,n+1)}$ is closed under all small limits (that exists) in $\mathbf{Cat}_{(\infty,n+1)}$.

Proof: Let $F\colon I \to \mathbf{Cat}^{S\text{-adj}}_{(\infty,n+1)}$ be a diagram. For $k \in S$ and assume the limit of F exists in $\mathbf{Cat}_{(\infty,n+1)}$. Then by Corollary 2.1.1, we have a commutative diagram

$$\begin{tikzcd}
{[\mathbf{Adj}_k, \varprojlim_{i:\,\mathcal{I}} F(i)]} \arrow[r, hook] \arrow[d, "\simeq"'] & {[c_k, \varprojlim_{i:\,\mathcal{I}} F(i)]} \arrow[d, "\simeq"] \\
\varprojlim_{i:\,\mathcal{I}} [\mathbf{Adj}_k, F(i)] \arrow[r, "\simeq"'] & \varprojlim_{i:\,\mathcal{I}} [c_k, F(i)]
\end{tikzcd}$$

where the bottom horizontal map is an equivalence, which gives that the top horizontal map is an equivalence. This means that $\varprojlim F$ has S-adjoints. ■

Proposition 2.8 (Free and Sub-Categories with Adjoints)

Let $n \geq 0$ and $S \subset \{1, \ldots, n\}$. The inclusion $\mathbf{Cat}^{S\text{-adj}}_{(\infty,n+1)} \hookrightarrow \mathbf{Cat}_{(\infty,n+1)}$ fits into a bireflection between locally finitely-presentable categories:

$$\begin{tikzcd}
\mathbf{Cat}^{S\text{-adj}}_{(\infty,n+1)} \arrow[r, hook] \arrow[r, bend left=60, "\mathsf{Free}^S"] \arrow[r, bend right=60, "\mathsf{Sub}^S"'] \arrow[r, phantom, shift left=3, "\perp"] \arrow[r, phantom, shift right=3, "\perp"] & \mathbf{Cat}_{(\infty,n+1)}
\end{tikzcd}$$

Furthermore, the unit of the top adjunction $\mathcal{C} \twoheadrightarrow \mathsf{Free}^S(\mathcal{C})$ is an epimorphism and the counit of the bottom adjunction $\mathsf{Sub}^S(\mathcal{C}) \hookrightarrow \mathcal{C}$ is a monomorphism.

Proof: $\mathbf{Cat}_{(\infty,n+1)}$ has small colimits and is locally small. Clearly, $\mathbf{Cat}^{S\text{-adj}}_{(\infty,n+1)}$ is the category of $\mathcal{W}$-weakly local objects [Lur18, Def. 04LH] for the small collection $\mathcal{W} = \{c_k \overset{L}{\twoheadrightarrow} \mathbf{Adj}_k \mid k \in S\} \cup \{c_k \overset{R}{\twoheadrightarrow} \mathbf{Adj}_k \mid k \in S\}$. The domain of these maps of $\mathcal{W}$ are all c_k and hence compact. The codomain of these maps are all $\mathbf{Adj}_k$'s and hence compact by Corollary 2.6.1. By the Small Object Argument [Lur18, Cor. 04MD], the inclusion $\mathbf{Cat}^{S\text{-adj}}_{(\infty,n+1)} \hookrightarrow \mathbf{Cat}_{(\infty,n+1)}$ has a left adjoint Free^S.

Note that this imply $\mathbf{Cat}^{S\text{-adj}}_{(\infty,n+1)}$ has all colimits and that they're preserved by Free^S. Explicitly, these colimits are computed by taking the diagram in $\mathbf{Cat}_{(\infty,n+1)}$ via the inclusion, taking its colimit there, and then applying Free^S. By Lemma 2.6, the last step is not necessary: the colimit taken in $\mathbf{Cat}_{(\infty,n+1)}$ already has S-adjoints. Hence, the inclusion preserves all small colimits and thus is accessible. This shows that $\mathbf{Cat}^{S\text{-adj}}_{(\infty,n+1)}$ is a locally finitely-presentable category. By the Adjoint Functor Theorem [Lur18, Thm. 06Q5], the inclusion $\mathbf{Cat}^{S\text{-adj}}_{(\infty,n+1)} \hookrightarrow \mathbf{Cat}_{(\infty,n+1)}$ has a right adjoint Sub^S.

Because epimorphisms are stable under pushout and transfinite composition, the unit $\mathcal{C} \twoheadrightarrow \mathsf{Free}^S(\mathcal{C})$ is an epimorphism by construction. For $\mathcal{A}, \mathcal{B} \in \mathbf{Cat}_{(\infty,n+1)}$, we then have $[\mathcal{A}, \mathsf{Sub}^S(\mathcal{B})] \simeq [\mathsf{Free}^S(\mathcal{A}), \mathsf{Sub}^S(\mathcal{B})] \simeq [\mathsf{Free}^S(\mathcal{A}), \mathcal{B}] \hookrightarrow [\mathcal{A}, \mathcal{B}]$, showing that the counit $\mathsf{Sub}^S(\mathcal{B}) \hookrightarrow \mathcal{B}$ is a monomorphism. ■

Corollary 2.8.1

Let $S \subset K \subset \{1, \ldots, n\}$. The inclusion $\mathbf{Cat}^{K\text{-adj}}_{(\infty,n+1)} \hookrightarrow \mathbf{Cat}^{S\text{-adj}}_{(\infty,n+1)}$ fits into a bireflection between locally finitely-presentable categories:

$$\begin{tikzcd}
\mathbf{Cat}^{K\text{-adj}}_{(\infty,n+1)} \arrow[r, hook] \arrow[r, bend left=60, "\mathsf{Free}^{S\to K}"] \arrow[r, bend right=60, "\mathsf{Sub}^{S\to K}"'] \arrow[r, phantom, shift left=3, "\perp"] \arrow[r, phantom, shift right=3, "\perp"] & \mathbf{Cat}^{S\text{-adj}}_{(\infty,n+1)}
\end{tikzcd}$$

Furthermore, the unit of the top adjunction $\mathcal{C} \twoheadrightarrow \mathsf{Free}^{S\to K}(\mathcal{C})$ is an epimorphism and the counit of the bottom adjunction $\mathsf{Sub}^{S\to K}(\mathcal{C}) \hookrightarrow \mathcal{C}$ is a monomorphism.

Proof: The argument is entirely similar to the one of Lemma 2.8. We can identify $\mathbf{Cat}^{K\text{-adj}}_{(\infty,n+1)} \hookrightarrow \mathbf{Cat}^{S\text{-adj}}_{(\infty,n+1)}$ with the full subcategory of $\mathcal{W}$-weakly local objects for the small collection $\mathcal{W} = \{\mathsf{Free}^S(c_k \overset{L}{\twoheadrightarrow} \mathbf{Adj}_k) \mid k \in K\} \cup \{\mathsf{Free}^S(c_k \overset{R}{\twoheadrightarrow} \mathbf{Adj}_k) \mid k \in K\}$. The functor Free^S preserves compact objects (its right adjoint preserves filtered colimit) and epimorphisms (as it preserves pushouts), so we can apply the same argument as above. ■

Lemma 2.9 (Free preserves (some) Ajoints)

Let $S \subset \{2, \ldots, n\}$ and $m < \min(S)$. Then $\mathsf{Free}^{\emptyset\to S}$ sends $\mathbf{Cat}^{m\text{-adj}}_{(\infty,n+1)}$ to $\mathbf{Cat}^{S\cup\{m\}\text{-adj}}_{(\infty,n+1)}$.

Proof: Let $\mathcal{C} \in \mathbf{Cat}^{m\text{-adj}}_{(\infty,n+1)}$. Recall reflection $\mathcal{C} \twoheadrightarrow \mathsf{Free}^{\emptyset\to S}(\mathcal{C})$ is a transfinite composition of κ-many pushouts of maps in $\{c_k \xrightarrow{L} \mathbf{Adj}_k \mid k \in S\} \cup \{c_k \xrightarrow{R} \mathbf{Adj}_k \mid k \in S\}$, for some ordinal κ. We show by transfinite induction that the codomain of any such composition lands in $\mathbf{Cat}^{m\text{-adj}}_{(\infty,n+1)}$.

- **Initialization:** By definition, $\mathcal{C}_0 = \mathcal{C} \colon \mathbf{Cat}^{m\text{-adj}}_{(\infty,n+1)}$.
- **Successor step:** Let $\alpha < \kappa$. There is a $k \in S$ and a pushout diagram

$$\begin{tikzcd} c_k \arrow[r] \arrow[d] & \mathbf{Adj}_k \arrow[d, dashed] \\ \mathcal{C}_\alpha \arrow[r, dashed] \arrow[ur, phantom, "\ulcorner"] & \mathcal{C}_{\alpha+1} \end{tikzcd}$$

  By Lemma 2.5, it suffices to show the m-morphisms $c_m \to \mathbb{L}^{\Delta}_S(\mathcal{C}_\alpha \sqcup_{c_k} \mathbf{Adj}_k)$ have adjoints. It's clear that for $a_1, \ldots, a_m \geq 0$, we have $(c_k)_{a_1,\ldots,a_m,0,\ldots,0} \simeq (\mathbf{Adj}_k)_{a_1,\ldots,a_m,0,\ldots,0}$. Hence, we have an equivalence $(\mathcal{C}_\alpha)_{a_1,\ldots,a_m,0,\ldots,0} \simeq (\mathcal{C}_\alpha \sqcup_{c_k} \mathbf{Adj}_k)_{a_1,\ldots,a_m,0,\ldots,0}$. So the pushout is already Segal in the first m-directions and in the iterative Necklace formula [BS24, Thm. D], the first m steps do nothing. The last steps don't affect the space of m-morphisms either, so we have $(\mathcal{C}_\alpha)_{a_1,\ldots,a_m,0,\ldots,0} \simeq (\mathbb{L}^{\Delta}_S(\mathcal{C}_\alpha \sqcup_{c_k} \mathbf{Adj}_k))_{a_1,\ldots,a_m,0,\ldots,0}$. By assumption, every m-morphism in $\mathcal{C}_\alpha$ has an adjoint, which concludes the successor step.
- **Limit step:** Let $\lambda \leq \kappa$ be a limit ordinal. We can write $\mathcal{C}_\lambda = \underset{\longrightarrow}{\mathrm{colim}}_{\alpha<\lambda} \mathcal{C}_\alpha$, a colimit of categories with m-adjoints. By Lemma 2.7, we have $\mathcal{C}_\lambda \colon \mathbf{Cat}^{m\text{-adj}}_{(\infty,n+1)}$.

By the transfinite induction principle, we get $\mathsf{Free}^{\emptyset\to S}(\mathcal{C}) \colon \mathbf{Cat}^{m\text{-adj}}_{(\infty,n+1)} \cap \mathbf{Cat}^{S\text{-adj}}_{(\infty,n+1)} = \mathbf{Cat}^{S\cup\{m\}\text{-adj}}_{(\infty,n+1)}$. ■

Corollary 2.9.1 (Composing Free/Sub Constructions)

Let $k < n$ and $S \subset \{k+1, \ldots, n\}$. Then $\mathsf{Free}^{\{k\}\cup S} = \mathsf{Free}^S \circ \mathsf{Free}^{\{k\}}$.

Proof: Let $\mathcal{A} \in \mathbf{Cat}_{(\infty,n+1)}$. By Lemma 2.9, we know that $\mathsf{Free}^S \circ \mathsf{Free}^{\{k\}}(\mathcal{A})$ has $\{k\} \cup S$-adjoints. The prove the first clim, it suffices to see that it satifies the correct universal property. ■

2.3 Lifting Once-Fully-Adjoint Morphisms

Proposition 2.10 (Free Full Adjunction)

Denote $\mathbb{Z}^{\to\leftarrow}$ the poset whose objects are the integers and which has morphisms $2n \to 2n \pm 1$ for all $n \colon \mathbb{Z}$. We have a colimit diagram

$$\begin{tikzcd} & & & \mathsf{Free}^{\{k\}}(c_k) & & & \\ \cdots \arrow[urrr, dashed, two heads] & \mathbf{Adj}_k \arrow[ur, two heads] & & \mathbf{Adj}_k \arrow[u, dashed, two heads] & & \mathbf{Adj}_k \arrow[ul, two heads] & \cdots \arrow[ullll, dashed, two heads] \\ & c_k \arrow[ul, two heads, "R"] \arrow[ur, two heads, "L"'] & & c_k \arrow[ul, two heads, "R"] \arrow[ur, two heads, "L"'] & & c_k \arrow[ul, two heads, "R"] \arrow[ur, two heads, "L"'] & \\ & & & & & c_k \arrow[ul, two heads, "R"] \arrow[ur, two heads, "L"'] & \end{tikzcd}$$

We denote by $f^{\vee n} \colon c_k \to \mathsf{Free}^{\{k\}}(c_k)$ the n-th cocone projection (from left to right, mneaning that $\vee$ correspond to taking right adjoints), which are all epimorphisms.

Proof: It suffices to show the result for $k = 1$, as by Remark 2.2, suspension is the composition of a left adjoint and the forgetful from the slice, and hence commutes with contractible colimits. Firstly, we build the cocone diagram. By definition, we know that $\mathsf{Free}^{\{1\}-\mathsf{adj}}(c_1)$ is fully-$\{1\}$-adjoint and receives a universal functor $c_1 \to \mathsf{Free}^{\{1\}-\mathsf{adj}}(c_1)$. The space of cocones is the limit of the diagram $\mathbb{Z}^{\to\leftarrow} \to \mathcal{S}$ below:

$$\begin{tikzcd} \cdots \arrow[dr, "R^*"] & & [\mathbf{Adj}, \mathsf{Free}^{\{1\}-\mathsf{adj}}(c_1)] \arrow[dl, "L^*"'] \arrow[dr, "R^*"] & & \cdots \arrow[dl, "L^*"'] \\ & [c_1, \mathsf{Free}^{\{1\}-\mathsf{adj}}(c_1)] & & [c_1, \mathsf{Free}^{\{1\}-\mathsf{adj}}(c_1)] & \end{tikzcd}$$

By Theorem 2.1, this is an infinite zig-zag of weak homotopy equivalences. By picking the point in the zero-th space corresponding to the universal functor $c_1 \to \mathsf{Free}^{\{1\}-\mathsf{adj}}(c_1)$, we get a canonical cocone diagram.

Secondly, we build a functor $\mathsf{Free}^{\{1\}-\mathsf{adj}}(c_1) \to \underset{\longrightarrow}{\mathrm{colim}}(\mathbb{Z}^{\to\leftarrow} \to \mathbf{Cat}_{(\infty,2)})$. By universal property of $\mathsf{Free}^{\{1\}-\mathsf{adj}}(c_1)$, it's sufficient to show that $\underset{\longrightarrow}{\mathrm{colim}}(\mathbb{Z}^{\to\leftarrow} \to \mathbf{Cat}_{(\infty,2)})$ has $\{1\}$-adjoints and to pick a 1-morphism $c_1 \to \underset{\longrightarrow}{\mathrm{colim}}(\mathbb{Z}^{\to\leftarrow} \to$

$\mathbf{Cat}_{(\infty,2)}$). The first part is a consequence of **??** and Lemma 2.5, as any morphism coming from a copy of $\mathbf{Adj}$ is given by composition of L's and R's, which have adjoints. The second part follows by choosing the 1-cell $c_1 \to \varinjlim(\mathbb{Z}^{\to\leftarrow} \to \mathbf{Cat}_{(\infty,2)})$ given by the universal cocone projection on the zero-th copy of c_1.

It's now clear that the two functors $\mathsf{Free}^{\{1\}-\mathsf{adj}}(c_1) \to \varinjlim(\mathbb{Z}^{\to\leftarrow} \to \mathbf{Cat}_{(\infty,2)})$ and $\varinjlim(\mathbb{Z}^{\to\leftarrow} \to \mathbf{Cat}_{(\infty,2)}) \to \mathsf{Free}^{\{1\}-\mathsf{adj}}(c_1)$ are inverses to each other by examining the universal properties. ■

Remark 2.3. This result can be generalized to construct any $\mathsf{Free}^{\{k,\dots,n\}-\mathsf{adj}}(c_k)$ as a colimit. This however requires a more careful generalization of Lemma 2.6. One can even proof that for $k < n$, the colimit diagrams can be chosen finite, by carefully looking at the notion of redundancy of adjunction data (see Lemma 4.1 for an example of such a result).

Corollary 2.10.1 (Lifting Once-Fully-Dualizable Morphisms)

Let $1 \leq k < n$. Suppose we have a monomoprhism $\mathcal{A} \hookrightarrow \mathcal{B} \colon \mathbf{Cat}_{(\infty,n)}$, a map $\mathsf{Free}^{\{k\}}(c_k) \to \mathcal{B}$ such that all precomposition with $f^{\vee n} \colon c_k \to \mathsf{Free}^{\{k\}}(c_k)$ lifts to $\mathcal{A}$ as in the diagram below. Then, there exists a dashed lift if and only if, for all n, there exists the dotted lifts:

$$\begin{tikzcd}
 & \mathcal{A} \arrow[rr, hook] & & \mathcal{B} \\
c_{k+1} \arrow[ur, dotted, tail] \arrow[dr] & & & \\
 & c_k \arrow[uu, tail] \arrow[r, two heads] \arrow[uur, dashed, tail] & \mathsf{Free}^{\{k\}}(c_k) \arrow[uu] & \\
c_{k+1} \arrow[uur, dotted, tail] \arrow[ur] & & &
\end{tikzcd}$$

$$\epsilon(f^{\vee n}) \qquad f^{\vee n} \qquad \eta(f^{\vee n})$$

In other words: the fully-$\{k\}$-dualizable morphism lift if all of its units and counits lift.

2.4 Invertibility

Lemma 2.11 (Classifying Space of Adj)

The localization of $\mathbf{Adj}$ by freely inverting η and ϵ is the contractible category.

Proof: In $\mathbf{Cat}_{(\infty,n)}$, the terminal functor $c_1 \to \{\bullet\}$ can be seen as a localization or truncation. Indeed, using completeness, precompositing by this map induces the inclusion $[c_1, \mathcal{C}]_{\simeq} \hookrightarrow [c_1, \mathcal{C}]$ of the connected component of all invertible 1-morphisms (for example, see [Rez01, Thm. 6.2 and Prop. 6.4]). We have a factorization $c_1 \twoheadrightarrow \mathbf{Adj} \twoheadrightarrow \mathbf{Adj}[\eta^{-1}, \epsilon^{-1}] \to \{\bullet\}$ where the composition is epimorphic. This implies that $\mathbf{Adj}[\eta^{-1}, \epsilon^{-1}] \twoheadrightarrow \{\bullet\}$ is epimorphic as well and hence, by applying any $[-, \mathcal{C}]$, we get

$$[c_1, \mathcal{C}]_{\simeq} \hookrightarrow [\mathbf{Adj}[\eta^{-1}, \epsilon^{-1}], \mathcal{C}] \hookrightarrow [\mathbf{Adj}, \mathcal{C}] \hookrightarrow [c_1, \mathcal{C}]$$

To show that $\mathbf{Adj}[\eta^{-1}, \epsilon^{-1}] \twoheadrightarrow \{\bullet\}$ is an equivalence, it's sufficient to show that: given any adjunction $L \colon a \underset{\perp}{\rightleftarrows} b : R \colon \mathcal{C}$, if η and ϵ are invertible, then L is invertible. Invertibility is a property detected by truncation to the homotopy 1-category, where this is a classical result of ordinary category theory. ■

Lemma 2.12 (Classifying Space of Higher Walking Adjoints)

Let $1 \leq k < n$ and $S \subset \{k, \dots, n-1\}$. The terminal functor $\mathsf{Free}^S(c_k) \twoheadrightarrow c_{k-1}$ is an epimorphism and exhibits c_{k-1} as the localization at all i-morphisms for $i > k$, meaning that it is the reflection map of the localization $|-|_{>k-1} \colon \mathbf{Cat}_{(\infty,n)} \to \mathbf{Cat}_{(\infty,k-1)}$.

Proof: As above, the composition $c_k \twoheadrightarrow \mathsf{Free}^S(c_k) \to \{\bullet\}$ is epimorphic, implying the first part of the statement. We proceed by downward induction on $|S|$.

- **Base case $|S| = 1$:** If $k \notin S$ then $\mathsf{Free}^S(c_k) = c_k \twoheadrightarrow c_{k-1}$ and the result is a consequence of completeness. Otherwise, the truncation functor $|-|_{>k-1}$ being left adjoint will preserve the colimit in Lemma 2.10. The diagram being sent to the terminal $n \in \mathbb{Z}^{\to\leftarrow} \mapsto \{\bullet\}$ diagram, we get $|(\mathsf{Free}^{\{k\}}(c_k)|_{>k-1} \simeq \{\bullet\}$, as $\mathbb{Z}^{\to\leftarrow}$ is contractible (this can be seen by Quillen's Theorem A).
- **Inductive step:** Let $|S| > 1$. As above, one can assume without loss of generality that $S = \{k\} \sqcup S'$. Similarly to the argument of Lemma 2.11, we want to show that the inclusion $[c_k, \mathcal{C}]_{\simeq} \hookrightarrow [|\,\mathsf{Free}^S(c_k)|_{>k-1}, \mathcal{C}]$ is an equivalence. By Corollary 2.9.1, this is equivalent to showing $[c_k, \mathcal{C}]_{\simeq} \hookrightarrow [\mathsf{Free}^{\{k\}}(c_k), \mathsf{Sub}^{S'}(\mathcal{C})]$ ■

3 Dualizability Data as Factorization Homology

Throughout the rest of this paper, we fixe a symmetric monoidal $(\infty,1)$-category $\mathcal{V}$ whose tensor product commutes with sifted colimits in both variables.

Remark 3.1 (Potential Generalizations). All the results of this section hold more generally for $\mathcal{E}_n$-monoidal $(\infty,1)$-categories $\mathcal{V}$ whose monoïdal structure commutes with geometric realizations in each variable. We opted to treat only these $\mathcal{V}$ to simplify the exposition.

3.1 Factorization Homology of Stratified Spaces

This section follows the theory of factorization homology of conically smooth stratified spaces developed by Ayala, Francis, Rozenblyum and Tanaka in [AFT17b, AFR18b, AFT17a]. Conical smoothness will be implicit throughout this section, and we will simply refer to the spaces as stratified spaces.

Construction 3.2. Recall from [AFT17a, Const. 2.4] that for any stratified space X with a $\mathcal{B}$-structure, we have an operad $\mathcal{D}\mathsf{isk}(\mathcal{B})_{/X} \to \mathbf{Fin}_*$ together with a morphism of operads $\mathcal{D}\mathsf{isk}(\mathcal{B})_{/X} \to \mathcal{D}\mathsf{isk}(\mathcal{B})$ and a notion of "pre-coCartesian" morphisms. We denote $\mathcal{D}\mathsf{isk}_{/X} := \mathcal{D}\mathsf{isk}(\mathcal{B}\mathsf{sc})_{/X}$. In [AFT17a, Obv. 2.7], it is observed that if X is endowed with a $\mathcal{B}$-structure, then we have an equivalence of operads $\mathcal{D}\mathsf{isk}(\mathcal{B})_{/X} \simeq \mathcal{D}\mathsf{isk}_{/X}$ induced by the right fibration $\mathcal{B} \to \mathcal{B}\mathsf{sc}$, which also preserves the notion of pre-coCartesian morphisms. Informally, this operad can be described as follows:

- its objects are pairs $(n, (\bigsqcup_{1\le i\le k_j} B_{i,j} \hookrightarrow X)_{1\le j\le n})$ where $n \ge 0$ and $\bigsqcup_{1\le i\le k_j} B_{i,j} \hookrightarrow X$ is some embedding of a finite disjoint union of basics into X,
- the space of morphisms from $a = (n, (\bigsqcup_{1\le i\le k_j} B_{i,j} \hookrightarrow X)_{1\le j\le n})$ to $b = (m, (\bigsqcup_{1\le i\le l_j} B'_{i,j} \hookrightarrow X)_{1\le j\le m})$ is disjoint union over all maps $\phi\colon \mathbf{Fin}_*(\mathbf{n}^+, \mathbf{m}^+)$ of the limit in spaces

$$\begin{tikzcd}
\mathcal{D}\mathsf{isk}(\mathcal{B})_{/X}(a,b) \arrow[rr, dashed] \arrow[d, dashed] \arrow[dr, phantom, "\rightsquigarrow"] & & \{\bullet\} \arrow[d] \\
\mathsf{Emb}(\bigsqcup_{j\in\phi^{-1}(j')} \bigsqcup_{1\le i\le k_j} B_{i,j}, \bigsqcup_{1\le i\le l_{j'}} B'_{i,j'}) \arrow[r] & \prod_{j\in\phi^{-1}(j')} \mathsf{Emb}(\bigsqcup_{1\le i\le k_j} B_{i,j}, X) &
\end{tikzcd}$$

- a morphism is pre-coCartesian if and only if for each $1 \le j \le m$, the induced map $\bigsqcup_{j\in\phi^{-1}(j')} \bigsqcup_{1\le i\le k_j} B_{i,j} \to \bigsqcup_{1\le i\le l_{j'}} B'_{i,j'}$ is an equivalence in $\mathcal{D}\mathsf{isk}$.

For $\mathcal{V}$ a symmetric monoidal $(\infty,1)$-category, seen as a coCartesian fibration $\mathcal{V}^{\otimes} \to \mathbf{Fin}$, we denote $\mathbf{Alg}_X(\mathcal{V}) \hookrightarrow \mathbf{Cat}_{(\infty,1)/\mathbf{Fin}_*}(\mathcal{D}\mathsf{isk}_{/X}, \mathcal{V}^{\otimes})$ the full subcategory on the the functors which send pre-coCartesian morphisms to coCartesian ones. Note that this construction is functorial in X with respect to open embeddings, as it comes from the functorial slice construction. In particular, $\mathbf{Aut}(X)$ acts on the operad $\mathcal{D}\mathsf{isk}_{/X}$ by equivalences. Similarly, we get an operad $\mathcal{M}\mathrm{fld}_{/X}$ by removing the condition that the $B_{i,j}$ are basics. In some good sense, the inclusion $\mathcal{D}\mathsf{isk}_{/X} \hookrightarrow \mathcal{M}\mathrm{fld}_{/X}$ generates $\mathcal{M}\mathrm{fld}_{/X}$ under specific sifted colimits, which are called factorization homology.

Remark 3.3 (Closed Strata Inclusions). For $C \hookrightarrow X$ the inclusion of a closed union of strata, we have a canonical morphism of operads $\mathcal{D}\mathsf{isk}_{/X} \to \mathcal{D}\mathsf{isk}_{/C}$ which "forgets" the algebraic data corresponding to the other strata.

We recall two different ways to transfer algebraic structures between stratified spaces.

Proposition 3.1 (Pushforward [AFT17a, Lem. 2.24 and Thm. 2.25])

For $\pi\colon X \twoheadrightarrow Y$ be a weak constructible bundle of stratified spaces, we have a map of operads $\mathcal{D}\mathsf{isk}_{/Y} \to \mathcal{M}\mathrm{fld}_{/X}$ which induces a commutative diagram

$$\begin{tikzcd}
\mathbf{Alg}_X(\mathcal{V}) \arrow[rr, "\int_X"] \arrow[dr, "\pi_*"'] & & \mathcal{V} \\
& \mathbf{Alg}_Y(\mathcal{V}) \arrow[ur, "\int_Y"'] &
\end{tikzcd}$$

Proposition 3.2 (Refinement [AFT17a, Lem. 2.23])

For $r\colon X \dashrightarrow Y$ be a refinement of stratified spaces, we have a localization of operads $\mathcal{D}\mathsf{isk}_{/X} \to \mathcal{D}\mathsf{isk}_{/Y}$

which induces a commutative diagram

$$\begin{tikzcd} \mathbf{Alg}_Y(\mathcal{V}) \arrow[rr, "\int_X"] \arrow[dr, hook, "r^*"'] & & \mathcal{V} \\ & \mathbf{Alg}_X(\mathcal{V}) \arrow[ur, "\int_Y"'] & \end{tikzcd}$$

where the bottom left functor is fully-faithful.

Remark 3.4 (Commutative Diagrams). We have seen that one can transfer algebraic structures using either:

- open embeddings (by functoriality of the slice construction),
- closed strata inclusions (Remark 3.3),
- weak constructible bundles (Lemma 3.1),
- refinements (Lemma 3.2).

All of these functors are first defined on a 1-categorical level "on-the-nose", then extended to the corresponding ∞-operads using [AFT17b, Prop. 2.22]. As a consequence, any commutative diagram of such stratified maps will automatically induce a commutative diagram of the corresponding functors between $(\infty, 1)$-categories of algebras. This will be used extensively in the rest of this section.

Remark 3.5 (Retractions). Suppose we have a "retraction" $X \hookrightarrow Y \overset{\pi}{\twoheadrightarrow} X$ where the first map is a closed inclusion of strata and the second map is a constructible bundle, and where for all $x \in X$, $\pi\colon Y_x \to \{x\}$ is the projection onto a cone point. Then, it's clear by construction that we have an equivalence between categories of algebras $\mathbf{Alg}_X(\mathcal{V}) \simeq \mathbf{Alg}_Y(\mathcal{V})$ induced by the pushforward and the restriction.

3.2 Handlebodies as Duality Datum

We fix $n \geq 1$ throughout this section. We consider various stratifications of $\mathbb{R}^n$ which will "carry" the algebraic structure of the duality data, through factorization homology's pushforward (3.1) and refinement (3.2) constructions. We denote the coordinates of a point $x \in \mathbb{R}^n$ by $(x_1, \ldots, x_n)$. The reader should think of the ith-coordinate as the direction of composition of i-morphisms in an (∞, n)-category.

The first stratification is to be thought of as the stratification which "carries" the structure of an arbitrary k-morphism:

Definition 3.1 (Bar-Stratification)**.** For $0 \leq k \leq n$, we define $\mathbb{R}^n_{\frac{n}{k}}$ to be the stratification of $\mathbb{R}^n$ given by the following strata:

- for $1 \leq i \leq k$, the strata given by $x_1 = x_2 = \ldots = x_{i-1} = 0$, and $\pm x_i > 0$,
- the stratum given by $x_1 = x_2 = \ldots = x_{k-1} = x_k = 0$.

This is a total of $2k + 1$ strata (by convention, when $k = 0$, there is a single stratum: $\mathbb{R}^n$ itself).

Remark 3.6 (Algebras over Bar-Stratifications). If $k < n$, the data of an algebra $M \in \mathbf{Alg}_{/\mathbb{R}^n_{\frac{n}{k}}}(\mathcal{V})$ is precisely the data of a k-morphism in the $(\infty, n+1)$-category $\mathbf{Mor}_n(\mathcal{V})$. If $k = n$, any such algebra gives a n-morphism in $\mathbf{Mor}_n(\mathcal{V})$, together with a choice of a point (an $\mathcal{E}_0$-structure).

The (co)domain and composition constructions can be seen as the following operations. For the rest of the exposition, we fix:

- a smooth function $\rho\colon \mathbb{R} \to [0, \infty)$ which is strictly decreasing on $(-\infty, 0]$, such that $\forall n \geq 0, \rho^{(n)}(0) = 0$ and $\forall x \geq 0, \rho(x) = 0$. For example, one may use $\rho(x) = e^{1/x}$ for all $x < 0$ and $\rho(x) = 0$ for $x \geq 0$.
- an increasing diffeomorphism $\phi\colon \mathbb{R} \to \mathbb{R}_{>0}$.

It is important to note that the choice of these functions does not matter (there's a contractible choice), although a precise statement of this fact is not necessary for our purposes.

Definition 3.2 ((Co)Domain)**.** For $1 \leq k \leq n$, we have an open embedding $\mathsf{dom}\colon \mathbb{R}^n_{\frac{n}{k-1}} \hookrightarrow \mathbb{R}^n_{\frac{n}{k}}$ which sends

$$x_k \mapsto \phi(x_k)$$

This induces an equivalence $\mathbb{R}^n_{\overline{k-1}} \simeq \left(\mathbb{R}^n_{\overline{k}}\right)_{x_k>0}$. Similarly, one has $\mathsf{cod}\colon \mathbb{R}^n_{\overline{k-1}} \hookrightarrow \mathbb{R}^n_{\overline{k}}$ given by $x_k \mapsto -\phi(-x_k)$

Definition 3.3 (Composition)**.** For $1 \leq k \leq n$, we define $\mathbb{R}^n_{\overline{k \circ \overline{k}}}$ to be the stratification of $\mathbb{R}^n$ given by the following strata:

- for $1 \leq i < k$, the strata given by $x_1 = x_2 = \ldots = x_{i-1} = 0$, and $\pm x_i > 0$,
- the strata given by $x_1 = x_2 = \ldots = x_{k-1} = 0$ and $x_k < -1$, $x_k = -1$, $-1 < x_k < 1$, $x_k = 1$, $1 < x_k$.

This is a total of $2k+3$ strata. We have a constructible bundle $\mathbb{R}^n_{\overline{k \circ \overline{k}}} \overset{\circ}{\twoheadrightarrow} \mathbb{R}^n_{\overline{k}}$ which sends

$$x_k \mapsto x_k \left(\rho(1 - x_k^2) + \sum_{i=1}^{k-1} x_i^2 \right)$$

intuitively pinching the two $(n-k)$-dimensional strata.

The second stratification of interest is to be thought of as the stratification which "carries" the structure of some specific duality datum of an arbitrary k-morphism.

Definition 3.4 (Duality Data)**.** Let $d < n$ and a word $\sigma \in \{\epsilon, \eta\}^d$ of length d in the letters "ϵ, η" is called a d-duality datum. We denote $|\eta| = 1$ and $|\epsilon| = -1$.

Definition 3.5 (Handlebody-Stratification)**.** Let $1 + d \leq k + d \leq n$ and $\sigma = \sigma_{k+d-1} \ldots \sigma_k \in \{\epsilon, \eta\}^d$ be a d-duality datum. We define $\mathbb{R}^n_{\overline{k,\sigma}}$ to be the stratification of $\mathbb{R}^n$ given by the following strata:

- **Bar-strata:** for $1 \leq i \leq k-1$, the strata given by $x_1 = x_2 = \ldots = x_{i-1} = 0$, and $\pm x_i > 0$,
- **Handlebody-strata:** $x_1 = x_2 = \ldots = x_{k-1} = 0$, and
  (a) $x_{k+d} > \sum_{i=k}^{k+d-1} |\sigma_i| x_i^2$,
  (b) $x_{k+d} = \sum_{i=k}^{k+d-1} |\sigma_i| x_i^2$,
  (c) $x_{k+d} < \sum_{i=k}^{k+d-1} |\sigma_i| x_i^2$.

This is a total of $2k+1$ strata.

Note that the stratification above is isomorphic to the k-bar-stratification by sending x_{k+d} to $x_{k+d} - \sum_{i=k}^{k+d-1} |\sigma_i| x_i^2$, and by permuting the order of the coordinates. In particular, algebras over $\mathbb{R}^n_{\overline{k,\sigma}}$ are also (pointed) k-morphisms in $\mathbf{Mor}_n(\mathcal{V})$, although maybe in a less canonical way (as it depends on the choice of permutation).

Finally, the third stratification is to be thought of as the stratification which "carries" the understanding that σ-duality data of an arbitrary k-morphism is a $(k+d)$-morphism. Strangely enough, the author encourages the reader to skip this definition on a first reading, and to come back to it after Definition 3.8. Indeed, the stratification might seem quite complicated but it's simply the minimal one which makes the "full-pinch" map into a stratified one.

Definition 3.6 (Fully-Barred Handlebody-Stratification)**.** Let $1 + d \leq k + d \leq n$ and $\sigma = \sigma_{k+d-1} \ldots \sigma_k \in \{\epsilon, \eta\}^d$ be a d-duality data. We define $\mathbb{R}^n_{\overline{k,\overline{\sigma}}}$ to be the stratification of $\mathbb{R}^n$ given by the following strata:

- **Bar-strata:** for $1 \leq i \leq k-1$, the strata given by $x_1 = x_2 = \ldots = x_{i-1} = 0$, and $\pm x_i > 0$,
- **Handlebody-Bar-strata:** Write $\{|\sigma| = 1\} = \{k \leq i < k+d \mid |\sigma_i| = 1\}$, and similarly for $\{|\sigma| = -1\}$. For $k \leq i < k+d$, write $\Delta_i = |\sigma_i| \left(x_{k+d} - \sum_{j=i}^{k+d-1} |\sigma_i| x_i^2 \right)$ with the convention $\Delta_{k+d} = -1$. For $k \leq j \leq k+d$, we have the following strata
  (a) the 2 strata given by $\pm x_j > 0$, $\Delta_j < 0$ and $x_u = 0$ for $u \in \{1, \ldots, j-1\}$,
  (b) for $k \leq i < j$, the 3 strata given by $x_u = 0$ for $u \in \{1, \ldots, k-1\} \sqcup \{|\sigma| = |\sigma_i|\} \cap \{k, \ldots, i-1\} \sqcup \{|\sigma| = -|\sigma_i|\} \cap \{k, \ldots, j-1\}$ and by either
     i. $\Delta_i > 0$, or
     ii. $\Delta_i = 0$ and $\pm x_i > 0$.

Finally, we have the stratum given by $x_1 = \ldots = x_{k+d} = 0$.

Note that we have an obvious refinement map $\mathbb{R}^n_{\overline{k},\overline{\sigma}} \dashrightarrow \mathbb{R}^n_{\overline{k},\sigma}$.

We now define some "pinch maps" which will help formalize the idea that the σ-duality data of an arbitrary k-morphism is a $(k+d)$-morphism.

Definition 3.7 (Last Coordinate Pinch)**.** As above, let $1+d \leq k+d \leq n$ and $\sigma = \sigma_{k+d-1} \ldots \sigma_k \in \{\epsilon, \eta\}^d$ be a d-duality data. We define a stratified map $\pi_k \colon \mathbb{R}^n_{\overline{k},\overline{\sigma}} \to \mathbb{R}^n_{\overline{k+1},\overline{\sigma_{k+d-1}\ldots\sigma_{k+1}}}$ called the "pinch map" by the following:

- it leaves unchanged all coordinates except the kth-one,
- it sends the kth-coordinate x_k to

$$x_k \longmapsto x_k \left(\rho \left(|\sigma_k| \left(x_{k+d} - \sum_{i=k}^{k+d-1} |\sigma_i| x_i^2 \right) \right) + \sum_{i=1}^{k-1} x_i^2 \right)$$

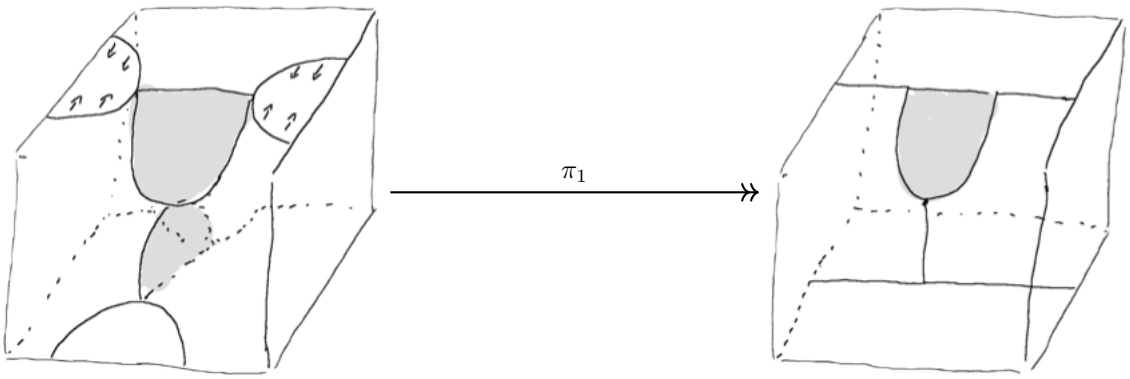

Figure 1: Pinching $\mathbb{R}^3_{\overline{1},\overline{-1,1}} \xrightarrow{\pi_1} \mathbb{R}^3_{\overline{1},\overline{1}}$. The small arrows indicate the collapsing that occurs "inside" the handle.

While we only need the easer notion of weak constructible bundle for our purposes, one can easily see that these maps are constructible bundle.

Lemma 3.3

As above, let $1+d \leq k+d \leq n$ and $\sigma = \sigma_{k+d-1} \ldots \sigma_k \in \{\epsilon, \eta\}^d$ be a d-duality data. The pinch

$$\mathbb{R}^n_{\overline{k},\overline{\sigma}} \xrightarrow{\pi_k} \mathbb{R}^n_{\overline{k+1},\overline{\sigma_{k+d-1}\ldots\sigma_{k+1}}}$$

is a (weak) constructible bundles, i.e. a stratified fiber bundle when corestricted to each strata of the codomain.

Proof: Consider a stratum of the codomain $\mathbb{R}^n_{\overline{k+1},\overline{\sigma_{k+d-1}\ldots\sigma_{k+1}}}$. We have to check that the preimage of the pinch map is a stratified fiber bundle over it. To simplify notations, we write $\overline{x}_k$ for the new k^{th}-coordinate after applying the pinch map. We denote the strata using a "path" notation indcating the order of the choices.

- **Bar-strata:** Clearly the bar-strata given by $i.\pm$ are unchanged for $i \leq k-1$. The preimage of the bar-strata $k.\pm$ are given by the handlebody-bar-strata $k.a.\pm$. For both these cases, the pinch map is diffeomorphic.
- **Handlebody-Bar-Strata:** Let $k+1 \leq j \leq k+d$. If $\sigma_k = \sigma_j$, the preimage of the handlebody-bar-strata $j.a.\pm$ are given by the handlebody-bar-strata $j.a.\pm$, and the pinch is diffeomorphic. If $\sigma_k \neq \sigma_j$, the preimage of the handlebody-bar-strata $j.a.\pm$ are given by the union of the three handlebody-bar-strata $j.k._$, where the pinch is pointwise given by a map equivalent to $[-1,1] \to \{0\}$. On these strata, one can check that the pinch is a fiber bundle. For $k+1 \leq i \leq j$, the preimage of the handlebody-bar-strata $j.i._$ are given by the corresponding handlebody-bar-strata $j.i._$, where the pinch is diffeomorphic.
- **Bottom-Bar-Stratum:** Corestricted to the stratum given by $x_1 = \ldots = x_{k+d} = 0$, the pinch map is clearly the identity, and hence a fiber bundle. ∎

Definition 3.8 (Full Pinch)**.** As above, let $1+d \leq k+d \leq n$ and $\sigma = \sigma_{k+d-1} \dots \sigma_k \in \{\epsilon, \eta\}^d$ be a d-duality data. We define the full pinch as the compositions

$$\mathbb{R}^n_{\overline{k},\overline{\sigma}} \twoheadrightarrow \mathbb{R}^n_{\overline{k+1},\overline{\sigma_{k+d-1}\dots\sigma_{k+1}}} \twoheadrightarrow \dots \twoheadrightarrow \mathbb{R}^n_{\overline{k+d-1},\overline{\sigma_{k+d-1}}} \twoheadrightarrow \mathbb{R}^n_{\overline{k+d},\overline{\emptyset}} = \mathbb{R}^n_{\overline{k+d}}$$

We define $\sigma\colon \mathbf{Alg}_{/\mathbb{R}^n_{\overline{k}}}(\mathcal{V}) \to \mathbf{Alg}_{/\mathbb{R}^n_{\overline{k+d}}}(\mathcal{V})$ to be the composition of the functors induced by:

$$\mathbb{R}^n_{\overline{k}} \xrightarrow{\sim} \mathbb{R}^n_{\overline{k},\sigma} \dashleftarrow \mathbb{R}^n_{\overline{k},\overline{\sigma}} \twoheadrightarrow \mathbb{R}^n_{\overline{k+d}}$$

Corollary 3.3.1

As above, let $1+d \leq k+d \leq n$ and $\sigma \in \{\epsilon, \eta\}^d$ be a d-duality data. The full pinch $\mathbb{R}^n_{\overline{k},\overline{\sigma}} \to \mathbb{R}^n_{\overline{k+d}}$ is a (weak) constructible bundle, i.e. a fiber bundle when restricted to each strata of the codomain.

Proof: This is a direct consequence of [AFR18b, Prop. 6.1.8]. ■

We briefly check that, for $d = 1$, the notation $\eta(f)$ and $\epsilon(f)$ introduced above do indeed exhibit an adjunction in the higher Morita category.

Lemma 3.4 (Duality Data Forms Adjunctions)

Let $k < n$ and $f \in \mathbf{Alg}_{/\mathbb{R}^n_{\overline{k}}}(\mathcal{V})$ be a k-morphism in $\mathbf{Mor}_n(\mathcal{V})$. Let f° be the k-morphism induced by $x_k \mapsto -x_k$ on $\mathbb{R}^n_{\overline{k}}$. Then, the $(k+1)$-morphisms $\eta(f)$ and $\epsilon(f)$ are respectively the unit and counit of an adjunction $f \dashv f^\circ$ in $\mathbf{Mor}_n(\mathcal{V})$.

Proof: By inspection, $\mathsf{cod}(\eta(f)) \simeq f^\circ \circ f$ and $\mathsf{dom}(\epsilon(f)) \simeq f \circ f^\circ$ using Definition 3.2 and Definition 3.3. The other domain and codomain can similalry be identitfied. The triangle identities then follow from the usual graphical calculus for adjunctions. ■

Note that, by construction, $(f^\circ)^\circ \simeq f$ and $\eta(f^\circ) \simeq \epsilon(f)^\circ$. We then have a direct corollary of Lemma 2.10.

Corollary 3.4.1 (Units in Morita)

Let $k < n$ and $f \in \mathbf{Alg}_{/\mathbb{R}^n_{\overline{k}}}(\mathcal{V})$ be a k-morphism in $\mathbf{Mor}_n(\mathcal{V})$. We have a lift $c_k \twoheadrightarrow \mathsf{Free}^{\{k\}}(c_k) \to \mathbf{Mor}_n(\mathcal{V})$ where the all induced (co)units are given by either of the following four algebras: $\eta(f)$, $\epsilon(f)$, $\eta(f^\circ)$, $\epsilon(f^\circ)$.

We now prove an associativity lemma that can be understood as "$\sigma(\eta(M)) \simeq \sigma\eta(M)$" (resp. for ϵ).

Lemma 3.5 (Left Associativity)

Let $1+d \leq k+d \leq n$ with $d \geq 2$ and let $\sigma = \sigma_{k+d-1} \dots \sigma_k \in \{\epsilon, \eta\}^d$ be a d-duality data. Define $\sigma' = \sigma_{k+d-1} \dots \sigma_{k+2} \in \{\epsilon, \eta\}^{d-1}$. We have a commutative diagram

$$\begin{tikzcd}
 & \mathbf{Alg}_{/\mathbb{R}^n_{\overline{k+1}}}(\mathcal{V}) \arrow[dr, "\sigma'"] & \\
\mathbf{Alg}_{/\mathbb{R}^n_{\overline{k}}}(\mathcal{V}) \arrow[ur, "\sigma_{k+1}"] \arrow[rr, "\sigma"'] & & \mathbf{Alg}_{/\mathbb{R}^n_{\overline{k+d}}}(\mathcal{V})
\end{tikzcd}$$

Proof: One can define a "partially-bar" handlebody stratification $\mathbb{R}^n_{\overline{k},\overline{\sigma_k},\sigma'}$, refining $\mathbb{R}^n_{\overline{k},\sigma}$ and minimal such that the pinch map $\mathbb{R}^n_{\overline{k},\overline{\sigma_k},\sigma'} \xrightarrow{\pi_k} \mathbb{R}^n_{\overline{k+1},\sigma'}$ is a map of stratified spaces. We then have a commutative diagram of stratified spaces

$$\begin{tikzcd}
\mathbb{R}^n_{\overline{k}} \arrow[r, "\sim"] \arrow[d, wavy] & \mathbb{R}^n_{\overline{k},\sigma_{k+1}} & \mathbb{R}^n_{\overline{k},\overline{\sigma_{k+1}}} \arrow[l, dashed] \arrow[r, two heads] \arrow[d, wavy] & \mathbb{R}^n_{\overline{k+1}} \arrow[d, wavy] \\
\mathbb{R}^n_{\overline{k},\sigma} \arrow[rr, dashed] & & \mathbb{R}^n_{\overline{k},\overline{\sigma_{k+1}},\sigma'} \arrow[ll, dashed] \arrow[r, two heads] & \mathbb{R}^n_{\overline{k+1},\sigma'} \\
\mathbb{R}^n_{\overline{k},\overline{\sigma}} \arrow[u, dashed] \arrow[urr, dashed] \arrow[rrr, two heads] & & & \mathbb{R}^n_{\overline{k+1},\overline{\sigma'}} \arrow[u, dashed] \arrow[r, two heads] & \mathbb{R}^n_{\overline{k+d}}
\end{tikzcd}$$

where the dashed arrows are refinements, the wavy ones are stratified equivalences and the $\twoheadrightarrow$ are (weak) constructible bundles given by the pinching. The outer top composition induces $\sigma' \circ \sigma_{k+1}$, while the outer bottom composition induces σ. ■

3.3 Dualizability of Iterated Bimodules

We mimick here an argument from [Lur17, Prop. 4.6.2.13] to identify dualizable iterated bimodules, using a similar idea to [Ram24, Cor. 3.8].

Proposition 3.6 (Dualizability for Iterated Bimodules)

Let M be a $\mathbb{R}^n_{\overline{n}}$-algebra in $\mathcal{V}$, seen as an n-morphism in $\mathbf{Mor}_n(\mathcal{V})$. Then, we have

- *M is a left adjoint in $\mathbf{Mor}_n(\mathcal{V})$ if and only if the induced $\mathbb{R}_{\geq 0}$-algebra, seen as a 1-morphism to the trivial algebra in $\mathbf{Mor}_1(\mathcal{V})$, is a left adjoint,*
- *M is a right adjoint in $\mathbf{Mor}_n(\mathcal{V})$ if and only if the induced $\mathbb{R}_{\leq 0}$-algebra, seen as a 1-morphism from the trivial algebra in $\mathbf{Mor}_1(\mathcal{V})$, is a right adjoint.*

Proof: The second statement clearly follows from the first by the "n-op" operation $x_n \mapsto -x_n$. Consider the following composition of constructible bundles $\mathbb{R}^n_{\overline{n}} \twoheadrightarrow \left(\mathbb{R}^2_{\overline{2}}\right)_{r\geq 0} \twoheadrightarrow \mathbb{R}^1_{\overline{1}}$, where the coordinates of the middle copy of $\mathbb{R}^2$ are denoted (x_n, r), defined by

$$(x_1, \ldots, x_n) \longmapsto \left(x_n, \sqrt{\sum_{i=1}^{n-1} x_i^2}\right) \longmapsto x_n$$

Using the iterated construction of the Morita category [Hau23, Cor. 7.6], or the factorization homology construction of composition in $\mathbf{Mor}_n(\mathcal{V})$, we see that M is a left adjoint in $\mathbf{Mor}_n(\mathcal{V})$ if and only if it is a left adjoint as a 1-morphism in $\mathbf{Mor}_1(\mathbf{Mod}_W(\mathcal{V}))$ where W is the $\mathcal{E}_2$-algebra supported on $r > 0$ in the pushforward $(x_n, r)^*(M)$. By [Lur17, Prop. 4.6.2.13], this is equivalent to the induced $\mathbb{R}_{\geq 0}$-algebra being a left adjoint in $\mathbf{Mor}_1(\mathbf{Mod}_W(\mathcal{V}))$. It remains to show that this is equivalent to being a left adjoint in $\mathbf{Mor}_1(\mathcal{V})$. One direction is clear by composition of adjunctions, so it suffices to show that being a left adjoint in $\mathbf{Mor}_1(\mathcal{V})$ implies being a left adjoint in $\mathbf{Mor}_1(\mathbf{Mod}_W(\mathcal{V}))$. The first part of the argument relies on a similar idea that the proof of [Lur17, Prop. 4.6.2.13], while the second part is a consequence of (a harmless generalization) of the argument in [Ram24, Cor. 3.8].

Let $\mathcal{W} = \mathbf{Mod}_W(\mathcal{V})$ and let B denote the induced $\mathbb{R}_{\geq 0}$-algebra. Note that $\mathbf{Mod}_B(\mathcal{V}) \simeq \mathbf{Mod}_B(\mathcal{W})$ by Remark 3.5, so we can abuse notation and denote both categories by $\mathbf{Mod}_B$. Consider the following lifitng problem:

$$\begin{tikzcd}
 & \mathcal{W} \arrow[dl, bend right, "\dashv"'] \arrow[dr, bend left, "\dashv"] & \\
\mathcal{V} \arrow[ur, bend left, "W\otimes -"] \arrow[rr, bend left, "B^\vee\otimes -"] & & \mathbf{Mod}_B \arrow[ul, bend right, "B\otimes -"'] \arrow[ll, bend left, "\perp", "B\otimes -"'] \arrow[ul, dashed, bend left, "\dashv"]
\end{tikzcd} \tag{3.1}$$

where we'd like to show the dotted left adjoint exists. We can consider the full subcategory $\mathcal{W}' \hookrightarrow \mathcal{W}$ on the objects w such that $\mathcal{W}(w, B\otimes -)\colon \mathbf{Mod}_B \to \mathcal{S}$ is corepresentable. Clearly, $\mathcal{W}'$ contains the free W-algebras (the image of the functor $\mathcal{V} \to \mathcal{W}$) and is closed under geometric realizations. Hence, $\mathcal{W}' = \mathcal{W}$ and the left adjoint exists.

Automatically, this left adjoint preserves colimits. To show that it is given by tensoring with some lift $B^\vee \in \mathcal{W}$, it suffices to check that this adjoint is a $\mathcal{W}$-module map. This follows from the argument in [Ram24, Cor. 3.8]. As a left adjoint to a module map, it is automatically an oplax-$\mathcal{W}$ map. Because $\mathcal{W}$ is generated under colimits by its unit W, it is in particular generated under colimits by dualizable objects. The "trace map" construction of [Ram24, Prop. 3.4] then shows that the oplax structure maps are equivalences. ■

Corollary 3.6.1 (Dualizability for Iterated Bimodules in Spectra)

Suppose $\mathcal{V} = \mathcal{S}p$. Let M be a $\mathbb{R}^n_{\overline{n}}$-algebra in $\mathcal{V}$, seen as an n-morphism in $\mathbf{Mor}_n(\mathcal{V})$. Then, we have

- *M is a left adjoint in $\mathbf{Mor}_n(\mathcal{V})$ if and only if it is compact as a module over the induced $\mathbb{R}_{>0}$-algebra, if and only if it is perfect as a module over the induced $\mathbb{R}_{>0}$-algebra,*
- *M is a right adjoint in $\mathbf{Mor}_n(\mathcal{V})$ if and only if it is compact as a module over the induced $\mathbb{R}_{<0}$-algebra, if and only if it is perfect as a module over the induced $\mathbb{R}_{<0}$-algebra,*

Proof: This is a direct consequence of [Lur17, 7.2.4.4]. ■

3.4 Centers for Dualizable Algebras

We recall here the content of [Lur17, Sec. 5.3.1] and detail how it applies to our setting.

Lemma 3.7 (Units as Maps to Centers)

Let $(\mathcal{A}, \mathcal{M}) \in \mathbf{Alg}_{\mathbb{R}_{\geq 0}}(\mathbf{Cat}_{(\infty,1)}^{\mathsf{geom}-\mathrm{colim}} \longrightarrow)$ and (A, M) a $\mathcal{D}\mathsf{isk}/\mathbb{R}_{\geq 0}$-algebra in $(\mathcal{A}, \mathcal{M})$. Suppose that we can extend the diagram bellow with the following dashed right adjoints.

$$\begin{tikzcd}
& \mathbf{Mod}_A(\mathcal{A}) \arrow[dl, bend right, "-\otimes A"'] \arrow[dr, bend left, "-\otimes_A M"] \arrow[dl, bend left, "\top"] \arrow[dr, dashed, bend right, "\hom_{\mathcal{M}}(M,-)"'] \arrow[dr, dashed, bend left, "\top"] & \\
\mathcal{A} \arrow[rr, bend left, "-\otimes M"] \arrow[rr, bend right, dashed, "\hom_{\mathcal{M}}(M,-)"'] \arrow[rr, phantom, "\perp"] & & \mathcal{M}
\end{tikzcd} \tag{3.2}$$

Then, when evaluating at $\mathbb{1}_{\mathcal{A}}$, the monad of the bottom adjuntion gives the center of M (seen as an object in $\mathcal{A}$) and the induced map of monads gives the universal $A \to \hom_{\mathcal{M}}(M, M)$ corresponding to the action of $A \curvearrowright M$ (seen as a morphism in $\mathcal{A}$).

Note that we only need and make this claim when seeing the map $A \to \hom_{\mathcal{M}}(M, M)$ as a map in $\mathcal{A}$. It's also true as a map in $\mathbf{Alg}_{\mathcal{E}^1}(\mathcal{A})$, using the induced $\mathcal{E}_1$-algebra structure on the map of monads, but we won't need this fact here.

Proof: Clearly, $\hom(M, M)$ is an endormorphism object for M [Lur17, Def. 4.2.1.28], and hence a center for M [Lur17, Lem. 5.3.1.11]. By the universal property of morphism objects, to identify maps $A \to \hom_{\mathcal{M}}(M, M)$, it suffices to identify the induced maps $A \otimes M \to M$. It's then clear by construction that the map of monads induces the correct action map on M. ■

Remark 3.7 (Lift of the Action of A on the Dual). One can actually prove the top dashed arrow exists if and only if the bottom dashed arrow exists. This is essentially contained in [Lur17, Sec. 5.3.1]. One way to see this is to observe that the left hand side adjunction is monadic and following the ideas of [Lur17, Sec. 4.7.3], we have: a (left) action of A on M gives a (right) action of the monad $-\otimes A$ on $-\otimes M$, which in turn is equivalent to a (left) action of the monad $-\otimes A$ on $\hom_{\mathcal{M}}(M, -)$, which in turn is equivalent to the existence of the dotted lift.

To complete the picture, we propose the following approach to re-prove the results of [Lur17, Sec. 5.3.1]. Now suppose $(\mathcal{A}, \mathcal{M}) \in \mathbf{Alg}_{\mathbb{R}_{\leq 0}\times\mathbb{R}^n}(\mathbf{Cat}_{(\infty,1)}^{\mathsf{geom}-\mathrm{colim}} \longrightarrow)$ and (A, M) a $\mathcal{D}\mathsf{isk}_{/\mathbb{R}_{\leq 0}\times\mathbb{R}^n}$-algebra in $(\mathcal{A}, \mathcal{M})$. Consider a similar diagram, but replacing $\mathcal{M}$ with $\mathbf{Mod}_M^{\bullet}(\mathcal{M})$, and assume again the existence of the dashed right adjoints. It's clear that all the left adjoint functors are $\mathcal{E}_n$-monoidal. Then, as can also be seen by [HHLN24, Cor. C], all the right adjoint functors are canoncially lax-$\mathcal{E}_n$-monoidal and the diagram induces

$$\begin{tikzcd}
& \mathbf{Alg}_{\mathcal{E}^n}(\mathbf{Mod}_A(\mathcal{A})) \arrow[dl, bend right, "-\otimes A"'] \arrow[dr, bend left, "-\otimes_A M"] \arrow[dl, bend left, "\top"] \arrow[dr, dashed, bend right, "\hom_M^{\bullet}(M,-)"'] \arrow[dr, dashed, bend left, "\top"] & \\
\mathbf{Alg}_{\mathcal{E}^n}(\mathcal{A}) \arrow[rr, bend left, "-\otimes M"] \arrow[rr, bend right, "\hom_M^{\bullet}(M,-)"'] \arrow[rr, phantom, "\perp"] & & \mathbf{Alg}_{\mathcal{E}^n}(\mathbf{Mod}_M^{\bullet}(\mathcal{M})) \arrow[r, dash, "\sim"] & \mathbf{Alg}_{\mathcal{E}^n}(\mathcal{M})_{M\backslash}
\end{tikzcd}$$

Informally, the induced monad of the bottom adjunction, evaluated at the identity $\mathbb{1}_{\mathcal{A}}$, upgrades to $\mathbf{Alg}_{\mathcal{E}^1}(\mathbf{Alg}_{\mathcal{E}^n}(\mathcal{A})) \simeq \mathbf{Alg}_{\mathcal{E}^{n+1}}(\mathcal{A})$ and can be identified with the $\mathcal{E}_n$-center of M. The induced map of monad, evalutated at the identity $\mathbb{1}_{\mathcal{A}}$, can be identified with the universal $\mathcal{E}_{n+1}$-map to the center $A \to \hom_M^{\bullet}(M, M)$. Thankfully, we don't need here to access any of the $\mathcal{E}_{n+1}$-monoïdal structures of these maps, so the following result suffices for our purposes.

Lemma 3.8 (Units as Maps to $\mathcal{E}_n$-Centers)

As above, let $(\mathcal{A}, \mathcal{M}) \in \mathbf{Alg}_{\mathbb{R}_{\leq 0}\times\mathbb{R}^n}(\mathbf{Cat}_{(\infty,1)}^{\mathsf{geom}-\mathrm{colim}} \longrightarrow)$ and (A, M) a $\mathcal{D}\mathsf{isk}_{/\mathbb{R}_{\leq 0}\times\mathbb{R}^n}$-algebra in $(\mathcal{A}, \mathcal{M})$. Suppose

that we can extend the diagram bellow with the following dashed right adjoints.

$$\begin{tikzcd}
& \mathbf{Mod}_A(\mathcal{A}) \arrow[dl, bend right, "-\otimes A"'] \arrow[dr, bend left, "-\otimes_A M"] \arrow[dl, phantom, "\perp"] \arrow[dr, phantom, "\perp"] \arrow[dr, bend right, dashed, "\hom^{\bullet}_M(M,-)"'] \arrow[dr, phantom, "\perp"] & \\
\mathcal{A} \arrow[ur, bend left] \arrow[rr, bend left, "-\otimes M"] \arrow[rr, phantom, "\perp"] & & \mathbf{Mod}^{\bullet}_M(\mathcal{M}) \arrow[ul, bend left, dashed, "\hom^{\bullet}_M(M,-)"]
\end{tikzcd} \tag{3.3}$$

Then, when evaluating at $\mathbb{1}_{\mathcal{A}}$, the monad of the bottom adjuntion gives the $\mathcal{E}_n$-center of M (seen as an object in $\mathcal{A}$) and the induced map of monads gives the universal $A \to \hom_{\mathcal{M}}(M, M)$ corresponding to the action of $A \curvearrowright M$ (seen as a morphism in $\mathcal{A}$).

Proof: This is an immidiate consequence of Lemma 3.7 and of the proof of [Lur17, Thm. 5.3.1.30]. ■

In our case, the $-\otimes_A M$ functor on the top right side of the diagram can be defined explicity with factorization homology as follows:

Remark 3.8. Let $(A, M, B) \in \mathbf{Alg}_{/\mathbb{R}^{n+1}_{\overline{1}}}(\mathcal{V})$. We can consider $(\mathcal{A}, \mathcal{M}) = (\mathcal{V}, \mathbf{Mod}_B(\mathcal{V})$ above, which canonically inherits the correct algebra structure via the collapsing of the $x_1 \geq 0$ region $\mathbb{R}^{n+1}_{\overline{1}} \twoheadrightarrow \mathbb{R}_{\leq 0} \times \mathbb{R}^n$. The functor $\mathbf{Mod}_A(\mathcal{V}) \xrightarrow{-\otimes_A M} \mathbf{Mod}^{\bullet}_M(\mathbf{Mod}_B(\mathcal{V}))$ above is then given by the bimodule action induced by

$$\mathbb{R}^{n+1}_{\overline{1}} \xleftarrow{\text{cl.}} (\mathbb{R}_{\leq 0} \times \{0\}^n) \bigsqcup_{\{0\}^{n+1}} (\mathbb{R}_{\geq 0} \times \mathbb{R}^n) \twoheadrightarrow \mathbb{R}_{\overline{1}}$$

where the first map is a closed inclusion, and the second map is the constructible bundle induced by taking the radius in the $x_1 > 0$ region.

Using this, we can now prove the following:

Proposition 3.9 (Identifying Units with Universal Maps to Centers)

Let $k + d = n$ and $\sigma = \sigma_{k+d-1} \dots \sigma_k \in \{\epsilon, \eta\}^d$ be a d-duality data. Let $a = |\{k \leq i < k + d \mid \sigma_i = \epsilon\}|$ and $b = d - a$. Let M be a $\mathbb{R}^n_{\overline{k}}$-algebra in $\mathcal{V}$, seen as an k-morphism in $\mathbf{Mor}_n(\mathcal{V})$. Let A, B be respectively the induced $\mathbb{R}_{<0}$ and $\mathbb{R}_{>0}$-algebras, as in Lemma 3.6, applied to M seen as a $\mathbb{R}^k_{\overline{k}}$-algebra. Suppose that $\sigma(M)$, seen as a n-morphism in $\mathbf{Mor}_n(\mathcal{V})$, has a right adjoint $\sigma(M)^\vee$. Then, the unit of the adjunction can be identified with the image in $\mathcal{V}$ of the canonical $\mathcal{E}_{b+1}$-map

$$\int_{S^{a-1} \subset D^a} (A, M) \longrightarrow \mathbf{HC}^B_{\mathcal{E}^b}(M)$$

where $\mathbf{HC}^B_{\mathcal{E}^b}(M)$ is the $\mathcal{E}_b$-center with $(\mathcal{A}, \mathcal{M}) = (\mathcal{V}, \mathbf{Mod}_B(\mathcal{V}))$ above, and where the action is induced by

$$\begin{tikzcd}
& \mathbb{R}^n_{\overline{k}} \\
(\{0\}^{a+1} \times \mathbb{R}^b \subset \{0\} \times \mathbb{R}^d \subset \mathbb{R}_{\geq 0} \times \mathbb{R}^d) \bigsqcup_{\{0\}^{a+1} \times \mathbb{R}^b} (\mathbb{R}_{\leq 0} \times \{0\}^a \times \mathbb{R}^b) \arrow[ur, hook, "\text{cl}"] \arrow[r, two heads] & \mathbb{R}^{b+1}_{\overline{1}}
\end{tikzcd}$$

and where the top inclusion is induced by the inclusion of the last $d + 1$ coordinates $\mathbb{R}^{d+1}_{\overline{1}} \xhookrightarrow{\text{cl.}} \mathbb{R}^n_{\overline{k}}$.

Proof: Let $\overline{A}$ and $\overline{B}$ be the induced $\mathbb{R}_{\geq 0}$ and $\mathbb{R}_{\leq 0}$-algebras, as in Lemma 3.6, applied to $\sigma(M)$. One can dualize

the diagram (3.1) and use it to extend the diagram (3.2) to the following:

$$\begin{tikzcd}
\mathbf{Mod}_{\overline{A}} \arrow[dd, bend right, "-\otimes\overline{A}"'] \arrow[dd, bend left, "\hom_{\mathbf{Mod}_{\overline{B}}}(M,-)"] \arrow[dr, bend left, "-\otimes_{\overline{A}} M"] \arrow[dr, phantom, "\prec"] & \\
& \mathbf{Mod}_{\overline{B}} \arrow[ul, bend left] \arrow[dl, bend left, "-\otimes M"] \arrow[dl, phantom, "\perp"] \arrow[dd, bend left, "\hom_{\mathbf{Mod}_{\overline{B}}}(M,-)"] \arrow[dd, phantom, "\succ"] \\
\mathcal{W} \arrow[ur, bend left] \arrow[dd, bend right, "-\otimes W"'] \arrow[dd, bend left] & \\
& \mathcal{V} \arrow[uu, bend left, "-\otimes M"] \arrow[uu, bend right, "\hom_{\mathbf{Mod}_{\overline{B}}}(M,-)"']
\end{tikzcd}$$

Under the downward forgetful functor $\mathcal{W} \to \mathcal{V}$, the unit of the top adjunction evaluated at $\overline{A}$ is clearly equivalent to the induced map of monad of the top triangle, evaluated at $\mathbb{1}_{\mathcal{W}}$, which in turn is clearly equivalent to the induced map of monad of the outer triangle, evaluated at $\mathbb{1}_{\mathcal{V}}$. By the above discussion, this unit can be identified with the canonical map to some center.

It now suffices to identify the functor $\mathbf{Mod}_{\overline{A}} \xrightarrow{-\otimes_{\overline{A}} M} \mathbf{Mod}_{\overline{B}}$ with the one induced by the action described in Remark 3.8 (where the "n" there is our "b"). By constuction, the functor $-\otimes_{\overline{A}} M$ above corresponds to the bimodule structure induced by the restriction to the last coordinate $\mathbb{R}_{\overline{1}} \hookrightarrow \mathbb{R}^n_{\overline{n}}$:

$$\begin{tikzcd}
& & X \arrow[r, two heads] \arrow[d, hook, "\mathrm{cl}"] \arrow[dr, phantom, "\lrcorner", very near start] & \mathbb{R}_{\overline{1}} \arrow[d, hook, "\mathrm{cl}"] \\
\mathbb{R}^n_{\overline{k}} & \mathbb{R}^n_{\overline{k},\sigma} \arrow[l, "\sim"'] & \mathbb{R}^n_{\overline{k},\overline{\sigma}} \arrow[l, "\mathrm{ref}"'] \arrow[r, two heads] & \mathbb{R}^n_{\overline{n}}
\end{tikzcd}$$

By Definition 3.6, the top map $X \twoheadrightarrow \mathbb{R}_{\overline{1}}$ factorises as $X \simeq \mathbb{R}_{\leq 0} \times \mathbb{R}^a \times \{0\}^b \bigsqcup_{\{0\}^{d+1}} \mathbb{R}_{\geq 0} \times \{0\}^a \times \mathbb{R}^b \twoheadrightarrow \mathbb{R}_{\overline{1}}$, where the second map is the radius on each factor. The equivalence is given by some permutation of coordinates. We then have the following diagram (see the figure bellow for a visual interpretation):

$$\begin{tikzcd}
& & & \mathbb{R}^{b+1}_{\overline{1}} \\
& Y \arrow[urrr, two heads] & & \left(\mathbb{R}_{\leq 0} \times \{0\}^b\right) \bigsqcup_{\{0\}^{b+1}} \left(\mathbb{R}_{\geq 0} \times \mathbb{R}^b\right) \arrow[u, hook, "\mathrm{cl}"'] \arrow[d, two heads] \\
\mathbb{R}^d_{\overline{1}} \arrow[d, hook] & \mathbb{R}_{\leq 0} \times \mathbb{R}^a \times \{0\}^b \bigsqcup_{\{0\}^{d+1}} \mathbb{R}_{\geq 0} \times \{0\}^a \times \mathbb{R}^b \arrow[l, hook, "\mathrm{cl}"'] \arrow[u, hook, "\mathrm{cl}"] \arrow[r, "\sim"'] \arrow[urr, two heads] & X \arrow[r, two heads] \arrow[d, hook, "\mathrm{cl}"] & \mathbb{R}_{\overline{1}} \arrow[d, hook, "\mathrm{cl}"] \\
\mathbb{R}^n_{\overline{k}} & \mathbb{R}^n_{\overline{k},\sigma} \arrow[l, "\sim"'] & \mathbb{R}^n_{\overline{k},\overline{\sigma}} \arrow[l, "\mathrm{ref}"'] \arrow[r, two heads] & \mathbb{R}^n_{\overline{n}}
\end{tikzcd} \tag{3.4}$$

where $Y = (\{0\}^{a+1} \times \mathbb{R}^b \subset \{0\} \times \mathbb{R}^d \subset \mathbb{R}_{\geq 0} \times \mathbb{R}^d) \bigsqcup_{\{0\}^{a+1} \times \mathbb{R}^b} (\mathbb{R}_{\leq 0} \times \{0\}^a \times \mathbb{R}^b)$ is the intermediate space in the statement of the result. The left/top side corresponds to the action described in the statement. The right side corresponds to the construction of Remark 3.8. The bottom corresponds to the original description coming from Definition 3.6. ■

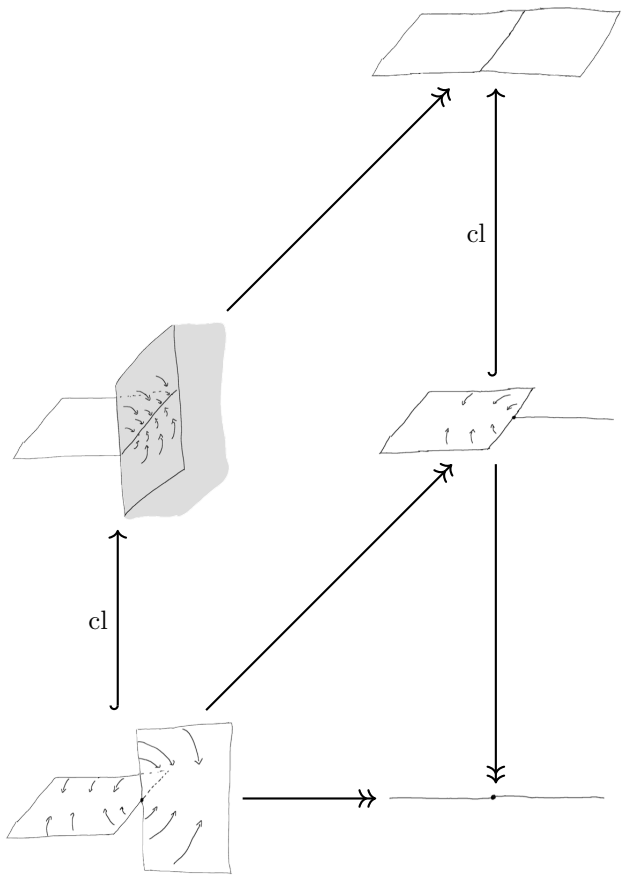

Figure 2: Visualization of the diagram (3.4).

4 Proof of the Criteria

4.1 Theorem A

The data exhibiting morphisms as higher full-adjoints is highly redundant. We show here an easy example of such redundancy that will be useful in the proof of the main theorem. This property is sometimes known as ambidexterity in the literature.

Lemma 4.1 (Redundancy of Duality)

Let $\mathcal{C} \in \mathbf{Cat}_{(\infty,3)}$ and suppose we have an adjunction $L\colon\ + \underset{\longleftarrow}{\overset{\longrightarrow}{\perp}} -\ : R$ such that the units and counits ϵ, η admit both left and right adjoints. Then ϵ and η are fully-adjoint.

Proof: Let $\epsilon^r \dashv \epsilon \dashv \epsilon^l$ and $\eta^r \dashv \eta \dashv \eta^l$. By taking left adjoints to each triangle iodentities, we can see that ϵ^l and η^l are respectively a unit and a counit for $R \dashv L$. Similarly, by taking right adjoints to each triangle identities, we can see that ϵ^r and η^r are respectively a unit and a counit for the same adjunction $R \dashv L$. By uniqueness of adjoints, we get $\epsilon^l \simeq \epsilon^r$ and $\eta^l \simeq \eta^r$, which then implies the result. ■

Theorem 4.2 (Full-Dualizability Criterion)

Let $\mathcal{V}\colon \mathbf{Alg}_{\mathcal{E}^\infty}(\mathbf{Cat}_{(\infty,1)})$ such that its tensor product commutes with sifted colimits in both variables. Let $0 \leq k < n$ and M be a k-morphism in $\mathbf{Mor}_n(\mathcal{V})$. We let A (resp. B) be the (co)domain of M if $1 < k$, or otherwise be the identity $\mathbb{1}_{\mathcal{V}}$ if $k = 0$. Then M is fully-dualizable if and only if, for all $1 \leq i \leq n-k$, the canonical $\mathcal{E}_1$-actions

$$\int_{(D^i, S^{i-1})} (A, M) \curvearrowright M \curvearrowleft \int_{(D^i, S^{i-1})} (B, M)$$

given by $\mathbb{R}^n_{\overline{k}} \hookleftarrow \{0\}^{k-1} \times \mathbb{R}_{\geq 0} \times \mathbb{R}^i \times \{0\}^{n-k-i} \twoheadrightarrow \mathbb{R}_{\geq 0}$ where the second map is the radius (resp. with $\leq$) (see Remark 3.4 for details on this construction), both exhibit M as a dualizable module.

Note that when $k = 0$, we recover the usual criterion first stated by Lurie [Lur09, Rem. 4.1.27]:

$$\forall i = 1, \ldots, n, \qquad M \text{ is a dualizable } \int_{S^{i-1}} M\text{-module.}$$

Proof: Let $d = n - k$. First, note that the actions described in the statement correspond precisely to all the actions involved in Lemma 3.6 applied to all of the $\sigma(M)$ and $\sigma(M^\circ)$ for $\sigma \in \{\epsilon, \delta\}^d$ a d-duality data. Indeed, one has a commutative diagram

$$\begin{tikzcd}
\mathbb{R}^n_{\overline{k}} \arrow[d, "(i)"] & \mathbb{R}_{\geq 0} \times \mathbb{R}^i \arrow[l, hook'] \arrow[r, two heads, "\mathrm{r}"] \arrow[d, hook, "(ii)"] & \mathbb{R}_{\geq 0} \arrow[d, hook] \\
\mathbb{R}^n_{\overline{k}} \arrow[r, "\sim"] & \mathbb{R}^n_{\overline{k},\sigma} & \mathbb{R}^n_{\overline{k},\overline{\sigma}} \arrow[l, dashed] \arrow[r, two heads] & \mathbb{R}^n_{\overline{n}}
\end{tikzcd}$$

where $a = |\{\sigma = \eta\}|$, (ii) is the inclusion of the coordinates $\{\sigma = \eta\} \subset \{k, \ldots, k+d-1\}$ and (i) is the obvious permutation of coordinates. One can pick (ii) such that the sign of the permutation is 1, showing that (i) is isotopic to the identity, which in turn exhibits the equivalence claimed between the module structures. Hence, it suffices to show by downward induction on k the following result: M is fully-dualizable if and only if, for all $\sigma \in \{\epsilon, \delta\}^d$ a d-duality data, $\sigma(M)$ and $\sigma(M^\circ)$ have both left and right adjoint, when seen as n-morphisms in $\mathbf{Mor}_n(\mathcal{V})$.

- **Case $k = n-1$:** By Lemma 4.1, the condition above is equivalent to having both left and right adjoints for each of the units and counits of the full-$\{n-1\}$-adjunction. By Corollary 3.4.1, these units and counits are described by the algebras $\eta(M)$, $\epsilon(M)$, $\eta(M^\circ)$, $\epsilon(M^\circ)$. By Lemma 3.6, this is equivalent to the condition in the statement of the theorem.
- **Inductive Step:** Note first that by Corollary 2.9.1 and Corollary 2.10.1, because $\mathbf{Mor}_n(\mathcal{V})$ has $\{1, \ldots, n-1\}$-adjoints, M is fully-dualizable if and only if we have a lift of the dashed arrow, if and only if we have a lift of each of the dotted arrows:

$$\begin{tikzcd}
 & \mathsf{Sub}^{\{k+1,\ldots,n\}}(\mathbf{Mor}_n(\mathcal{V})) \arrow[r, hook] & \mathbf{Mor}_n(\mathcal{V}) \\
c_{k+1} \arrow[ur, dotted] \arrow[dr, "\epsilon(f^{\vee n})"'] & & \\
 & \mathsf{Free}^{\{k\}}(c_k) \arrow[uu, dashed] \arrow[ur, dotted] & c_k \arrow[l, two heads] \arrow[uu, "M"'] \\
c_{k+1} \arrow[ur, "\eta(f^{\vee n})"'] & &
\end{tikzcd}$$

In other words, M is fully dualizable if and only if all the units and counits involved in the full-$\{k\}$-adjunction are themselves fully-dualizable. As above, by Corollary 3.4.1, these units and counits are described by the algebras $\eta(M)$, $\epsilon(M)$, $\eta(M^\circ)$, $\epsilon(M^\circ)$. By the inductive hypothesis, duality is then equivalent to: for all $\sigma \in \{\epsilon, \delta\}^{d-1}$ a $(d-1)$-duality data, $\sigma(\eta(M))$, $\sigma(\epsilon(M))$, $\sigma(\eta(M^\circ))$, $\sigma(\epsilon(M^\circ))$, $\sigma(\eta(M)^\circ)$, $\sigma(\epsilon(M)^\circ)$, $\sigma(\eta(M^\circ)^\circ)$ and $\sigma(\epsilon(M^\circ)^\circ)$ all have left and right adjoint, when seen as n-morphisms in $\mathbf{Mor}_n(\mathcal{V})$. Using Lemma 3.5 and the exchange property of $\circ$ with $\eta \leftrightarrow \epsilon$, we can conclude the proof. ■

Corollary 4.2.1 (Full-Dualizability of Ring Spectra)

Let $0 \leq k < n$ and M be a k-morphism in $\mathbf{Mor}_n(\mathcal{V})$. Then M is fully-dualizable if and only if, for all $1 \leq i \leq n-k$, the canonical $\mathcal{E}_1$-actions

$$\int_{(D^i, S^{i-1})} (A, M) \curvearrowright M \curvearrowleft \int_{(D^i, S^{i-1})} (B, M)$$

given by $\mathbb{R}^n_{\overline{k}} \hookleftarrow \{0\}^{k-1} \times \mathbb{R}_{\geq 0} \times \mathbb{R}^i \times \{0\}^{n-k-i} \twoheadrightarrow \mathbb{R}_{\geq 0}$ where the second map is the radius (resp. with $\leq$), both exhibit M as a perfect or compact module.

Proof: This is a direct consequence of Lemma 4.2 and [Lur17, Prop. 7.2.4.4]. ■

Remark 4.1 (Characterizing Full-Dualizability in Other Higher Categories). Implicitly, the proof uses that: if f is a fully-$\{k, \ldots, l\}$-dualizable k-morphism for $k \leq l$, full-$\{k, \ldots, l+1\}$-dualizability is equivalent to all $\sigma(f)$ having left adjoints for $\sigma \in \{\epsilon, \delta\}^{l-k+1}$ a $(l-k+1)$-duality data. This can be proven independently in any higher category with the tools of section 2 following the same argument. One just has to prove a bit more "redudancy" lemmas. We avoid this because, in our case, redundancy is baked into the Morita-cake using the actions $\mathbf{Aut}(X) \curvearrowright \mathbf{Alg}_{/X}$ given naturally by factorization homology.

Remark 4.2 (Other Proof Strategies). There are many possible proof strategies for this result, from closest to farthest from the one presented here:

- One could do a upward induction on k instead, using some right associativity lemma for the pinch constructions. This lemma is significantly heavier to spell out as the stratifications involved are more complicated, but the core idea is the same.
- One could also imagine a more elementary proof of the result above without explicit description of the pinch maps, by simply writing naïvely an inductive argument. This would however not translate well to the proof of Theorem B, where the explicit description of the pinch maps is very helpful to elucidate Lemma 3.9.
- One could also try to write the (∞, n)-functor $\mathcal{B}\mathrm{ord}_n^{\mathrm{fr}} \to \mathbf{Mor}_n(\mathcal{V})$ and use handlebody decomposition to identify which actions on the top n-morphism level need to be dualizable. This is a much more technical approach. Such a functor was constructed in [Sch14] for a pointed version of $\mathbf{Mor}_n(\mathcal{V})$ with the language of factorization algebras and relative categories. A proof following this approach would involve re-proving many technical results already present in [AFT17b, AFR18a, AFT17a] under a different formalism, or showing an equivalence of both formalisms. We avoid this by usng a crucial idea: deciding if a morphism is fully-dualizable is not necessarily a homotopy-coherent problem.

4.2 Theorem B

By analogy to the proof of Theorem A, we start with a redundancy lemma.

Lemma 4.3 (Redundancy of Invertibility)

Let $\mathcal{C} \in \mathbf{Cat}_{(\infty,2)}$ and suppose we have two parallel adjunctions

$$\begin{tikzcd} \bullet \arrow[r, "f", "\perp"'] & \bullet \arrow[l, bend right, "\perp"] \arrow[l, bend left, "\perp"] \end{tikzcd}$$

in $\mathcal{C}$ such that the units are each time equivalences. Then f is an equivalence.

Proof: Note that equivalences and adjoints are detected in the homotopy 2-category. Without loss of generality, one can then assume that $\mathcal{C}$ is a 2-category. In this case, one can use the 2-categorical Yoneda lemma [JY21] to reduce to the case where $\mathcal{C} = \mathbf{Cat}_{(\infty,1)}$. In $\mathbf{Cat}_{(1,1)}$, the unit of the bottom adjunction being a natural isomorphism implies that f is fully faithful, which in turn implies that the counit of the top adjunction is a natural isomorphism. Hence, the top adjunction exhibits f as an (adjoint) equivalence. ■

Theorem 4.4 (Invertibility Criterion)

Let $0 \leq k < n$ and $\mathcal{V} \in \mathbf{Alg}_{\mathcal{E}^\infty}(\mathbf{Cat}_{(\infty,1)})$. Let M be a k-morphism in $\mathbf{Mor}_n(\mathcal{V})$. Then M is invertible if and only if it is fully-dualizable and, for all $1 \leq i \leq n-k$, the universal maps

$$\int_{(D^i,S^{i-1})} (A, M) \xrightarrow{\sim} \mathbf{HC}^B_{\mathcal{E}^{i-1}}(M)$$
$$\int_{(D^i,S^{i-1})} (B, M) \xrightarrow{\sim} \mathbf{HC}^A_{\mathcal{E}^{i-1}}(M)$$

are invertible.

Proof: By Lemma 2.12, the functor $\mathsf{Free}^{\{k,\dots,n\}}(c_k) \twoheadrightarrow c_{k-1}$ is epimorphic and correspond under precomposition to some $[c_k, \mathbf{Mor}_n(\mathcal{V})]_\simeq \hookrightarrow [\mathsf{Free}^{\{k,\dots,n\}}(c_k), \mathbf{Mor}_n(\mathcal{V})]$. Hence, any invertible k-morphism is automatically fully-dualizable. We can then assume M is fully-dualizable, and show invertibility is equivalent to all these maps of algebra being invertible.

Let $d = n - k$ (which can be zero). Similarly to Theorem A, now using Lemma 3.9, we first observe that invertibility of all these maps is precisely equivalent to the statement: for all $\sigma \in \{\epsilon, \delta\}^d$ a d-duality data, the unit of $\sigma(M) \dashv \sigma(M)^\vee$ and of $\sigma(M)^\circ \dashv (\sigma(M)^\circ)^\vee$ are invertible. As a consequence, it suffices to show by downward induction on k the equivalence between invertibility of M and this condition.

- **Base Case $k = n$:** this is an immediate of consequence of the lemma above.
- **Inductive Step:** By Lemma 2.11 and Corollary 3.4.1, M is invertible if and only if $\epsilon(M)$ and $\eta(M)$ are invertible as $(k+1)$-morphisms in $\mathbf{Mor}_n(\mathcal{V})$. The statement then follows from the inductive hypothesis and Lemma 3.5. ■

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